



Chapter 4:

Motion in 2 dimensions

OBJECTIVES: Know and Apply

Definitions of Position, Velocity & Acceleration

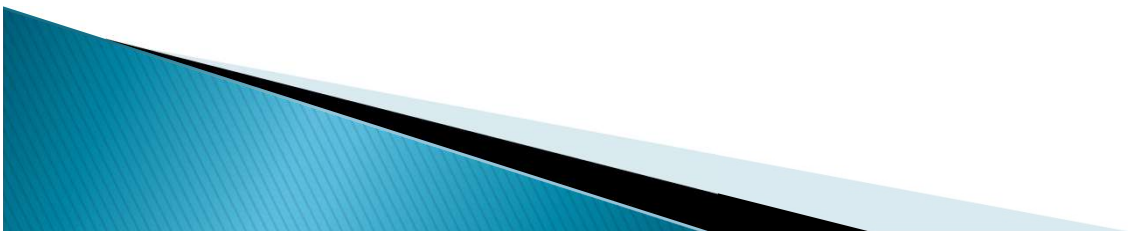
Describe Motion in a plane – 2D using vectors

Average Velocity, Average Acceleration

Projectile motion & Circular motion

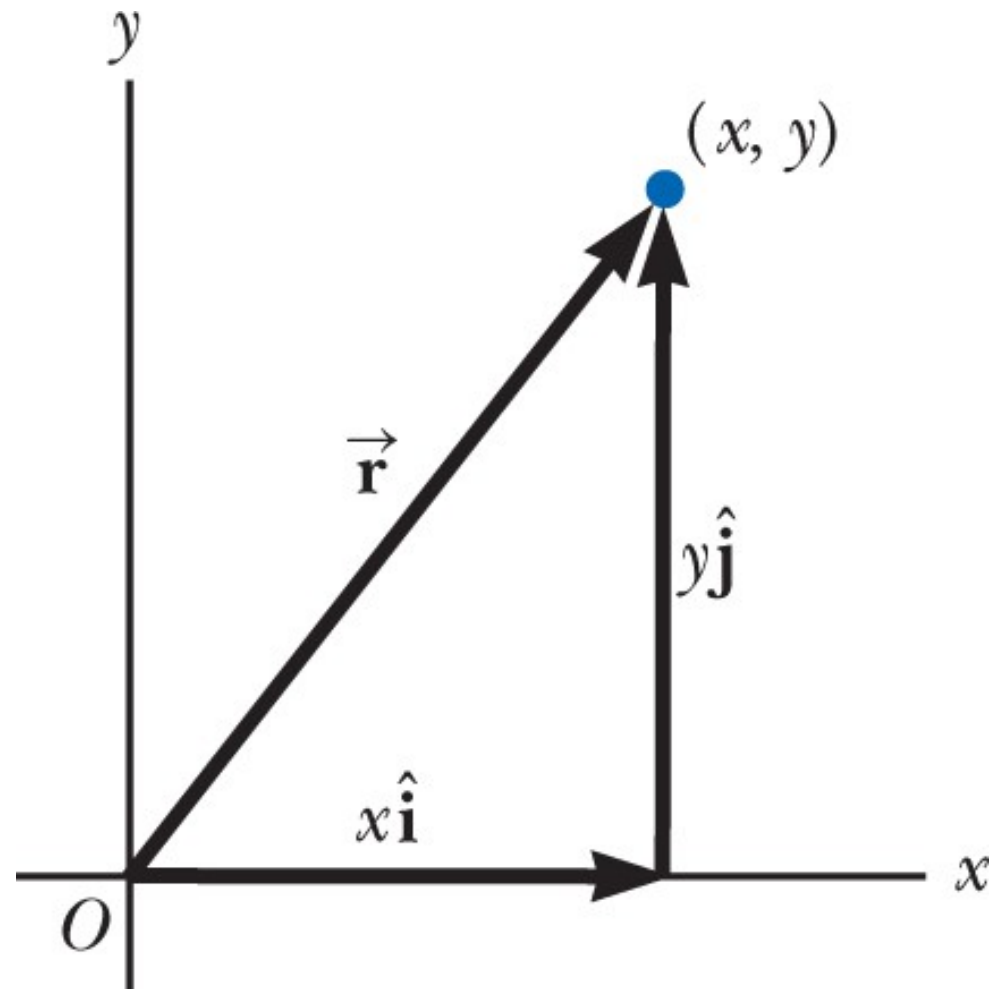
General Motion Ideas

- ▶ In two- or three-dimensional kinematics, everything is the same as in one-dimensional motion except that we must now use full vector notation.
 - Positive and negative signs are **no longer sufficient** to determine the direction.



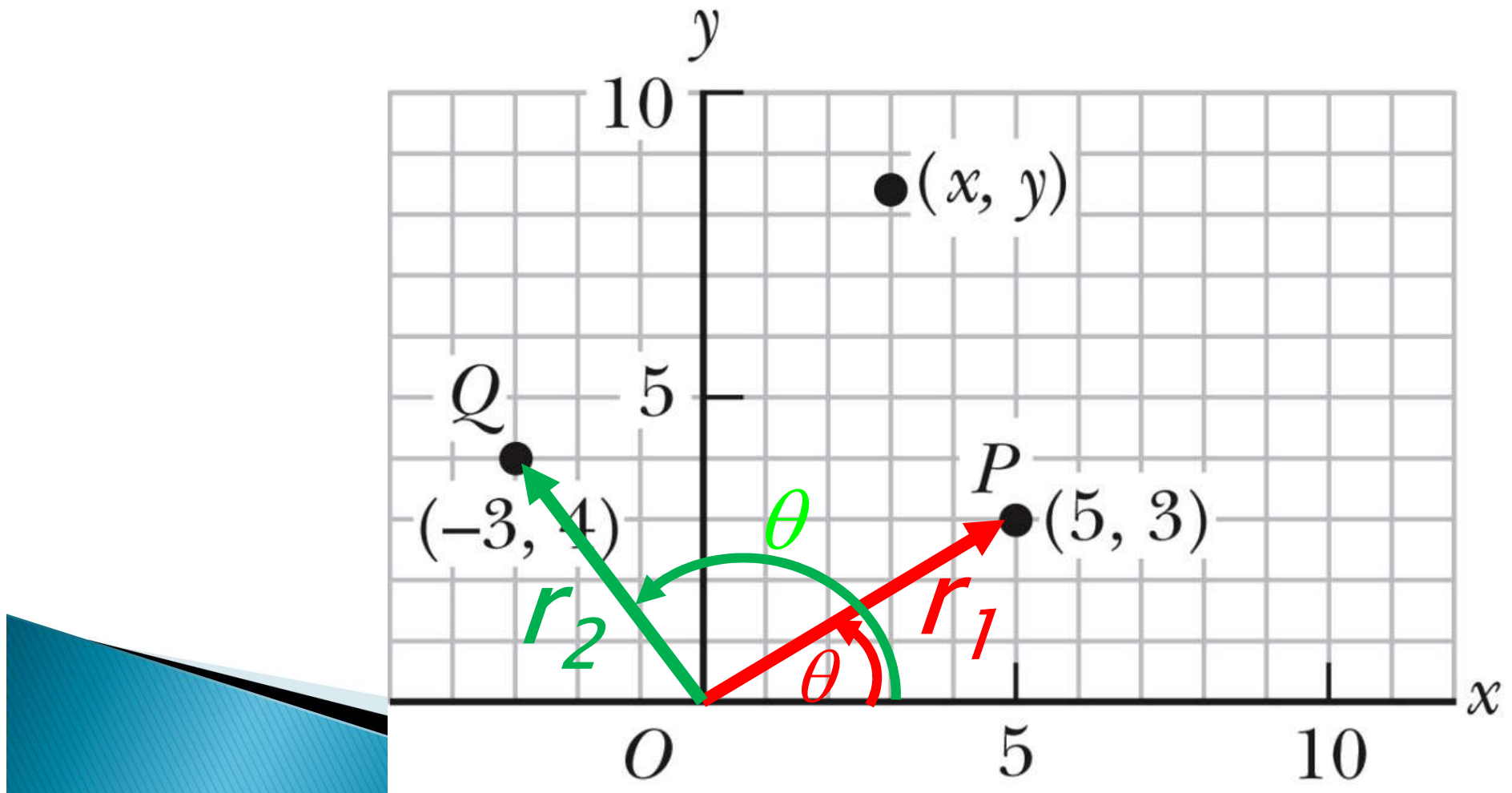
Position Vector

$$\vec{\mathbf{r}} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}}$$



Position Vector

A vector drawn from the origin of a reference frame / system of axes to the location of the point of interest



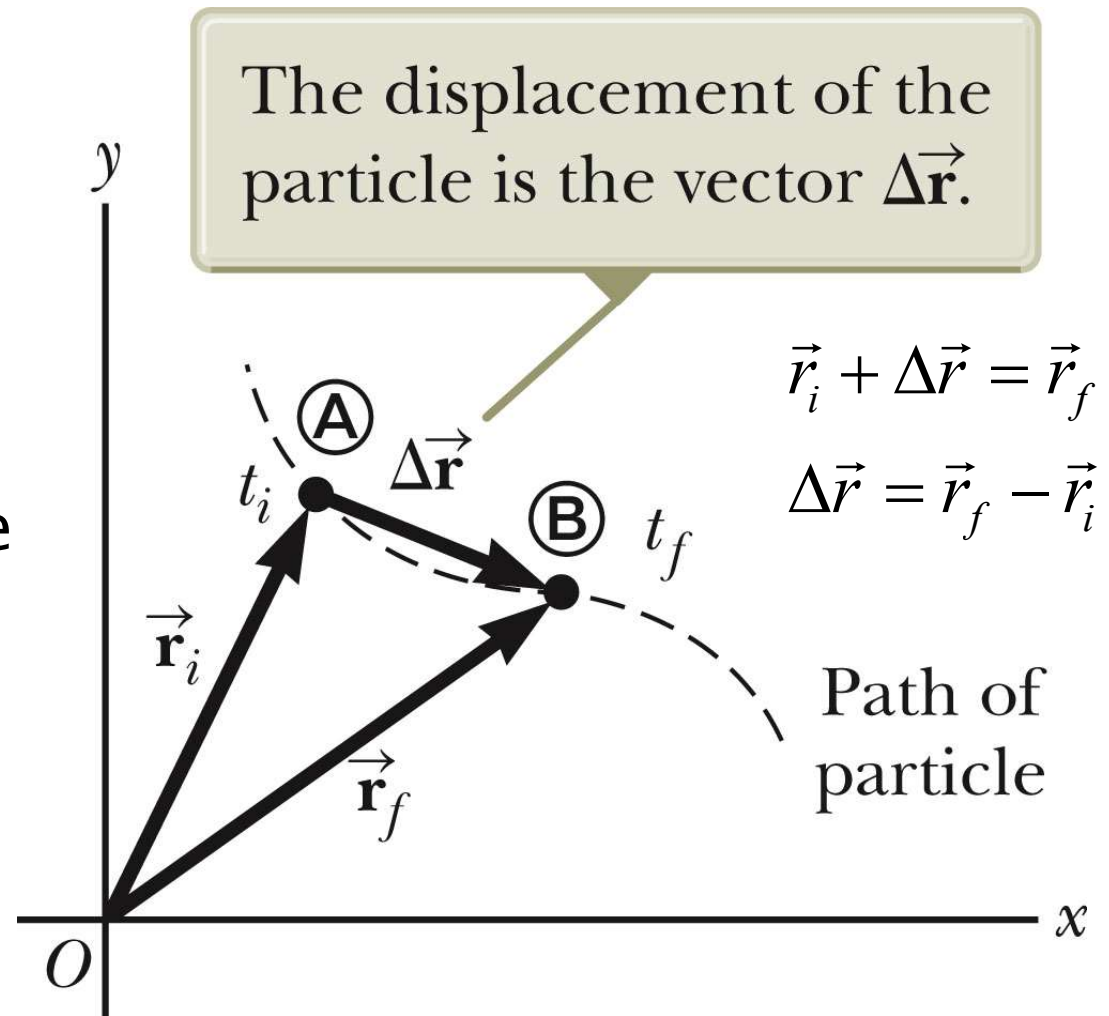
Position and Displacement Vectors

- ▶ The position of an object is described by its position vector, $\vec{r}$.
- ▶ The displacement of the object is defined as the change in its position.

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\Delta\vec{r} \equiv \vec{r}_f - \vec{r}_i$$

- ▶ Independent of path

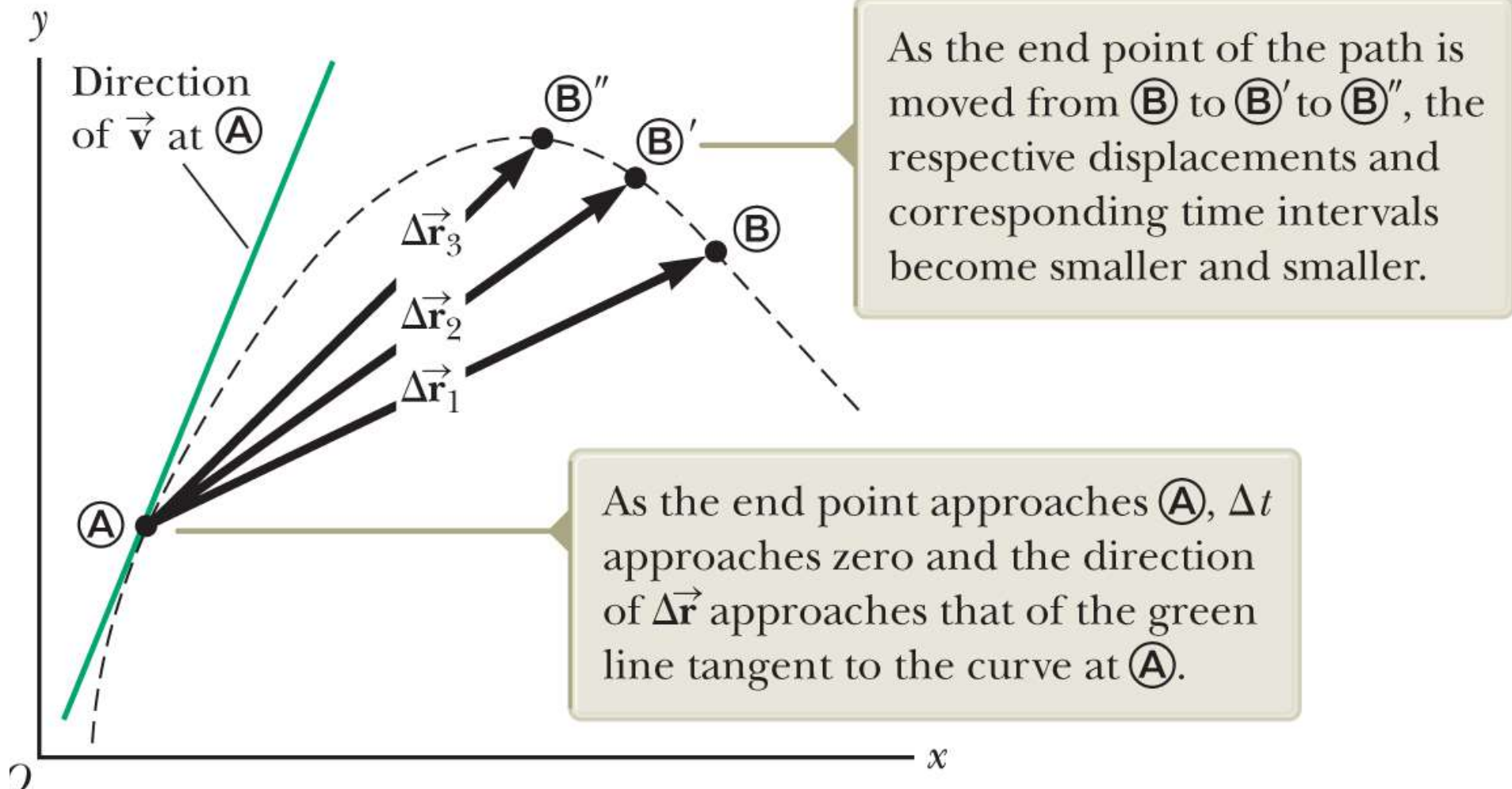


Velocity Vector

$$\vec{v}_{\text{avg}} \equiv \frac{\Delta \vec{r}}{\Delta t}$$

$$\vec{v} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$

$$v = |\vec{v}|$$



Average and Instantaneous Velocity

- Following the same approach as in Ch 2 we define average velocity:

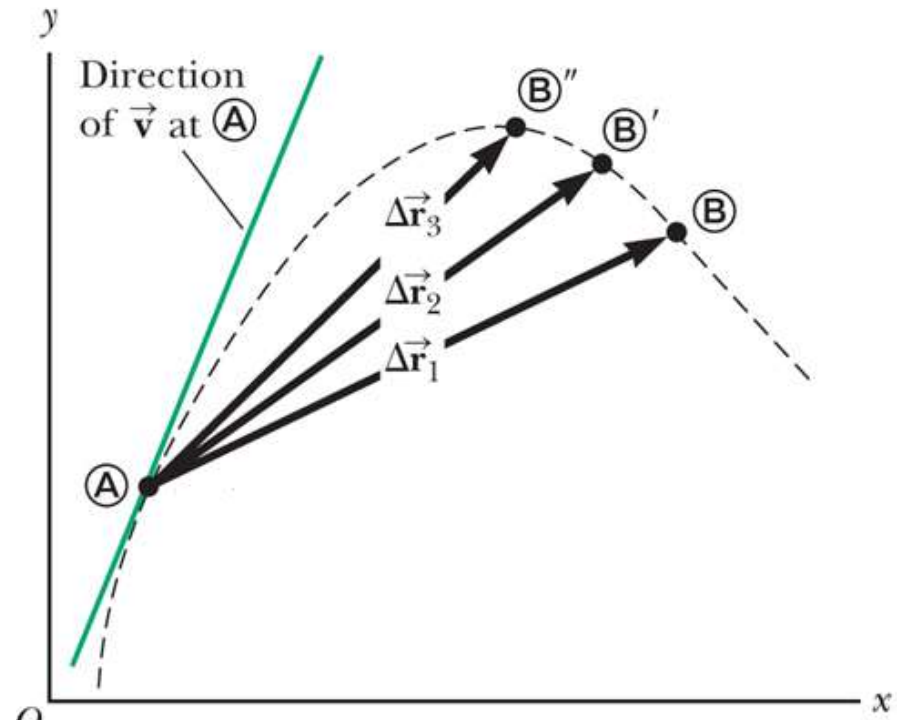
$$\vec{v}_{avg} \equiv \frac{\Delta \vec{r}}{\Delta t} = \frac{\Delta r_x \hat{i} + \Delta r_y \hat{j} + \Delta r_z \hat{k}}{\Delta t}$$

$$= \frac{\Delta r_x}{\Delta t} \hat{i} + \frac{\Delta r_y}{\Delta t} \hat{j} + \frac{\Delta r_z}{\Delta t} \hat{k}$$

$$\text{OR: } \frac{\Delta \vec{r}}{\Delta t} = \frac{\Delta x}{\Delta t} \hat{i} + \frac{\Delta y}{\Delta t} \hat{j} + \frac{\Delta z}{\Delta t} \hat{k}$$

- Instantaneous velocity given by:

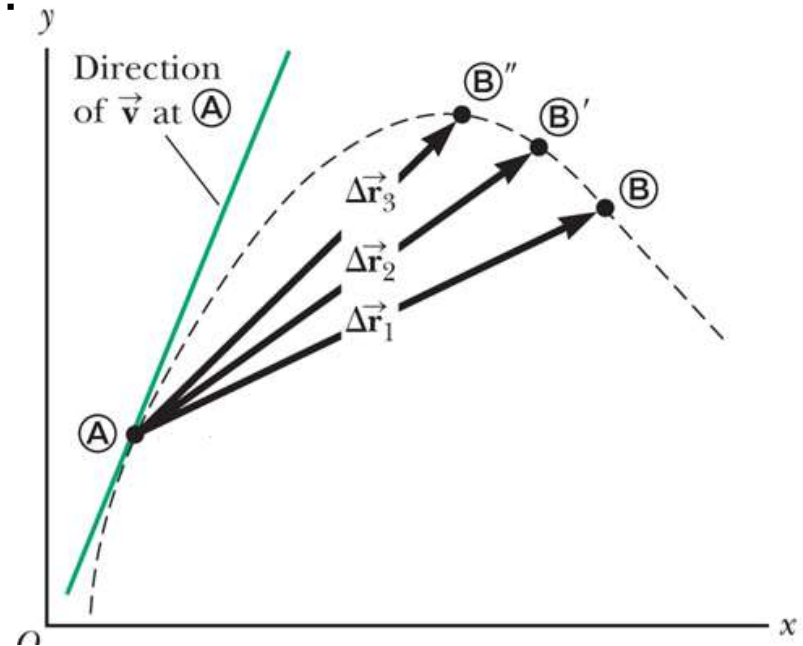
$$\vec{v} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$$



Average and Instantaneous Velocity

- ▶ Instantaneous velocity given by:

$$\begin{aligned}\vec{v} &\equiv \frac{d\vec{r}}{dt} = \frac{d(x\hat{i} + y\hat{j} + z\hat{k})}{dt} \\ &= \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k} \\ &= v_x\hat{i} + v_y\hat{j} + v_z\hat{k}\end{aligned}$$

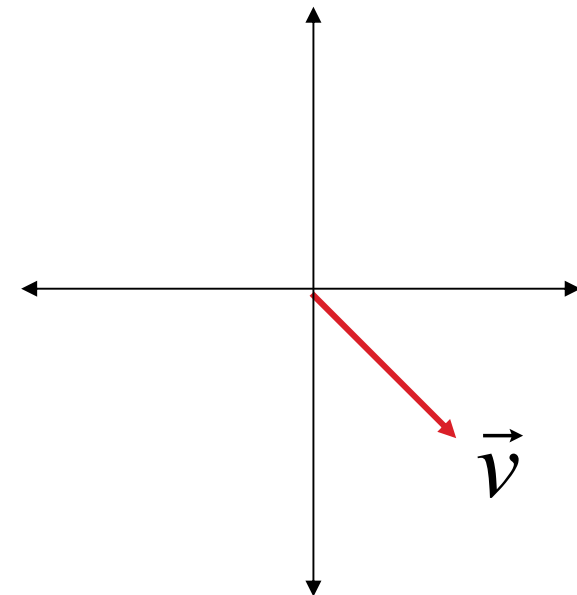
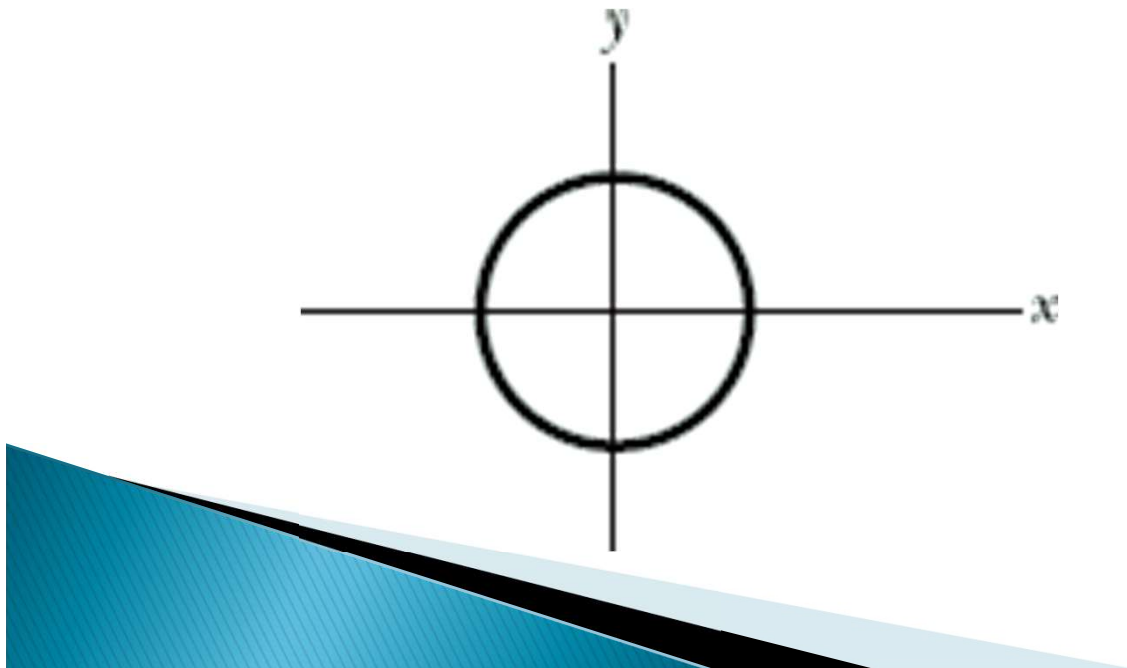


- ▶ Direction of $\vec{v}$ is always tangent to the particle's path.
- ▶ Scalar components of the velocity vector:

$$v_x = \frac{dx}{dt} \quad v_y = \frac{dy}{dt} \quad v_z = \frac{dz}{dt}$$

Example

The figure shows a circular path taken by a particle. If the instantaneous velocity of the particle is $\vec{v} = (2 \text{ m/s})\hat{i} - (2 \text{ m/s})\hat{j}$ through which quadrant is the particle moving at that instant if it is travelling (a) clockwise and (b) counter-clockwise around the circle? For both cases, draw on the figure.



Example

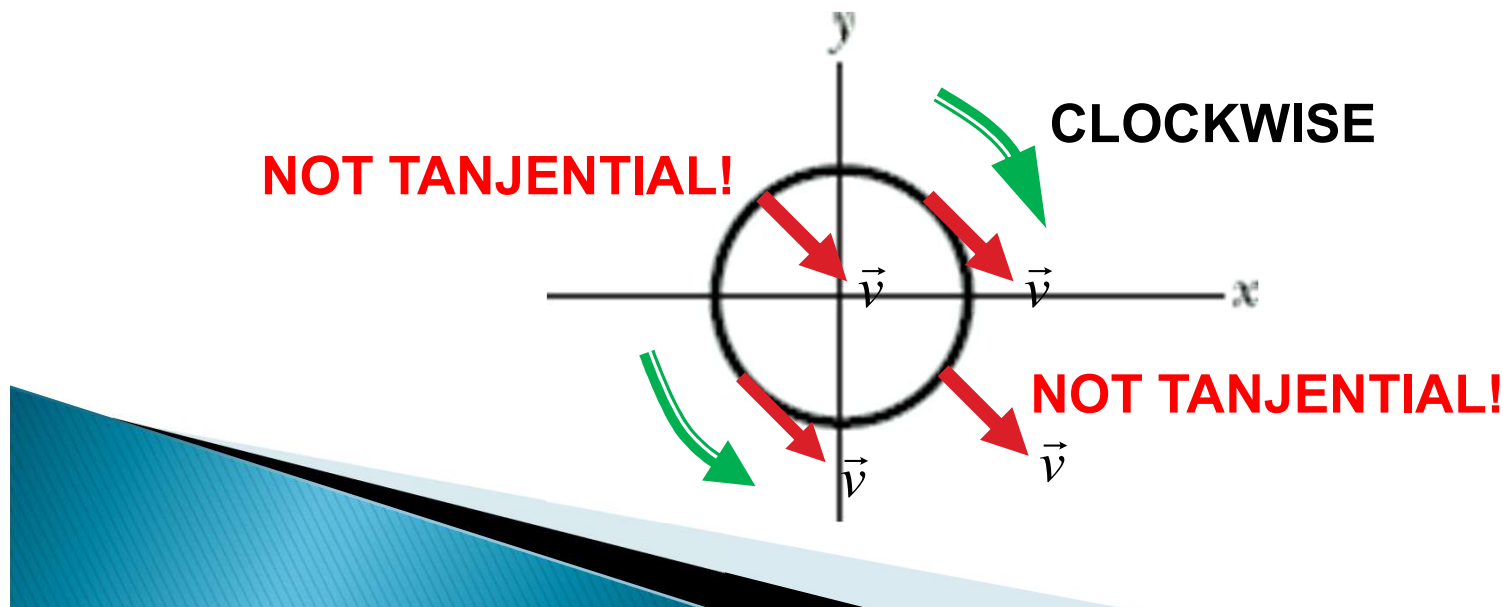
The figure shows a circular path taken by a particle. If the instantaneous velocity of the particle is:

$$\vec{v} = (2 \text{ m/s})\hat{i} - (2 \text{ m/s})\hat{j}$$

Through which quadrant is the particle moving at that instant if it is travelling (a) clockwise and (b) counter-clockwise around the circle? For both cases, draw on the figure.

Comment: Instantaneous velocity is tangential to the path!

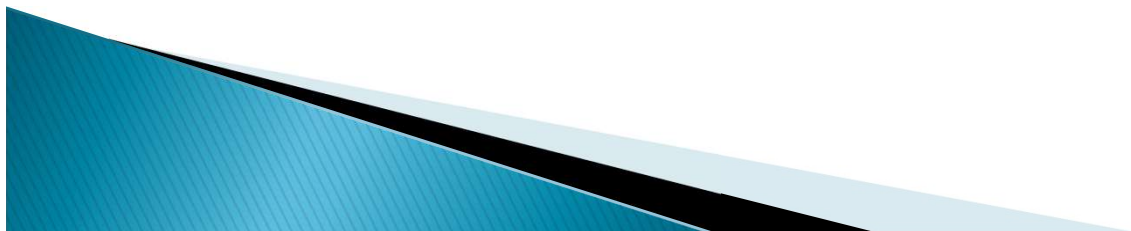
Q: Where can you draw a tangential velocity described by the $\vec{v}$?



Quick Quiz 4.1

Consider the following controls in an automobile in motion: accelerator pedal, brake, steering wheel. What are the controls in this list that cause an acceleration of the car?

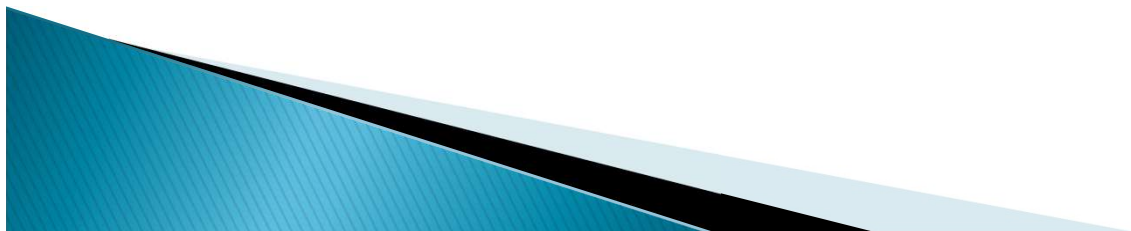
- (a) all three controls
- (b) the accelerator pedal and the brake
- (c) only the brake
- (d) only the accelerator pedal
- (e) only the steering wheel



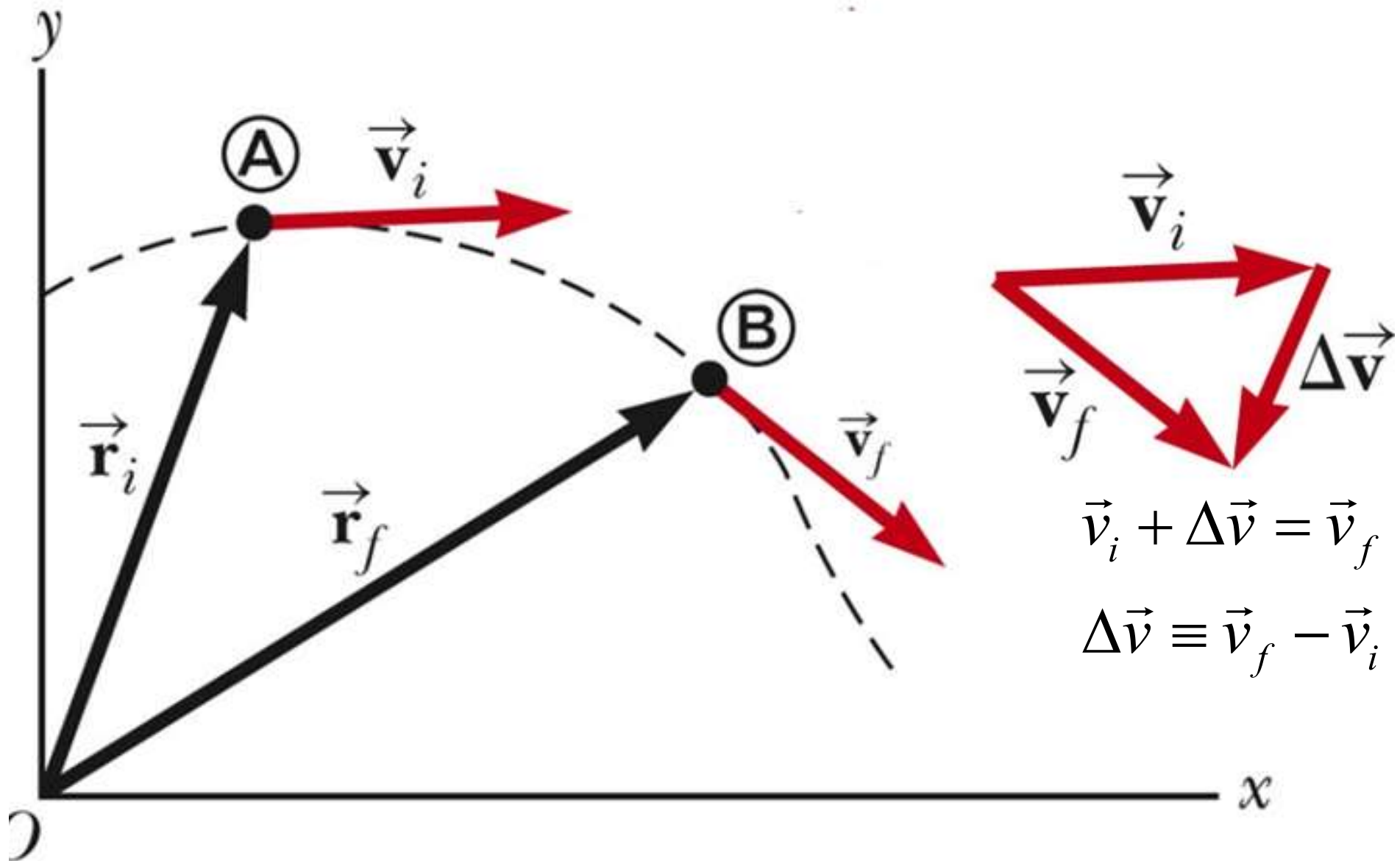
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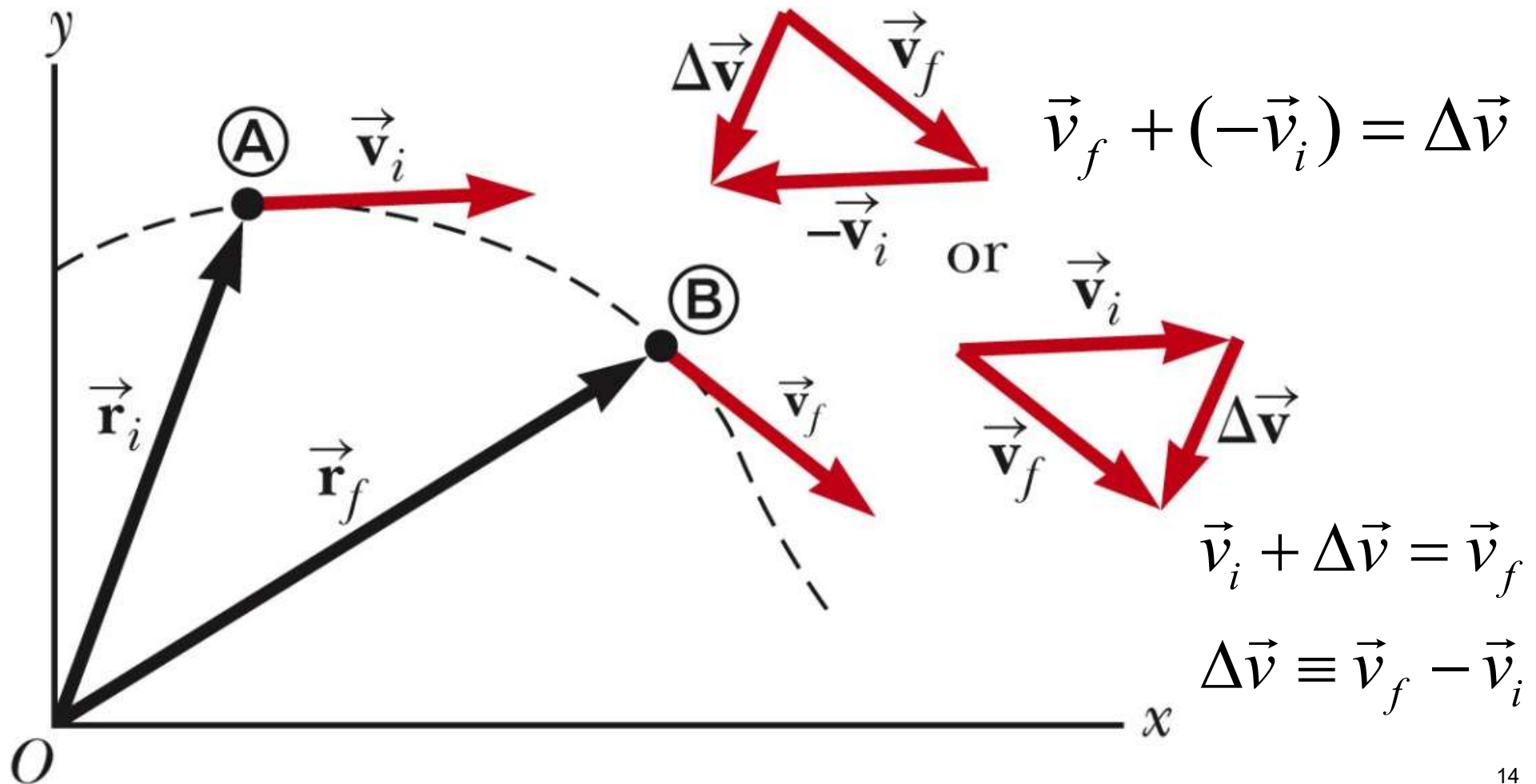
- (a) all three controls
- (b) the accelerator pedal and the brake
- (c) only the brake
- (d) only the accelerator pedal
- (e) only the steering wheel



Average and Instantaneous Acceleration



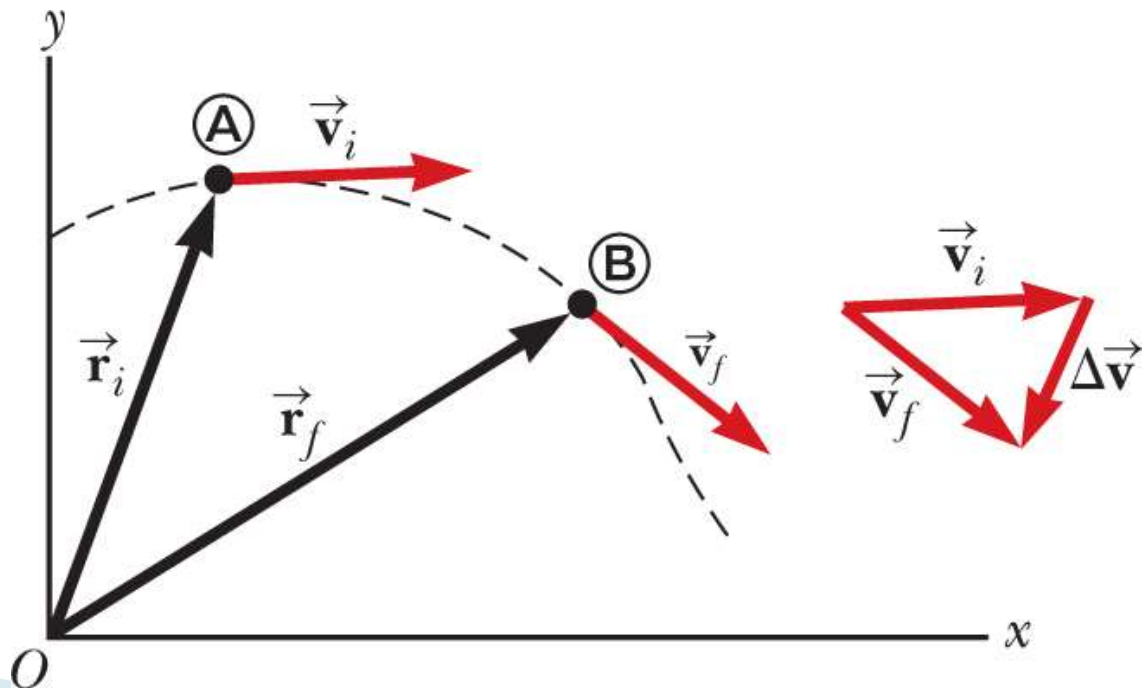
Average and Instantaneous Acceleration



Acceleration Vector

$$\vec{\mathbf{a}}_{\text{avg}} \equiv \frac{\Delta \vec{\mathbf{v}}}{\Delta t} = \frac{\vec{\mathbf{v}}_f - \vec{\mathbf{v}}_i}{t_f - t_i}$$

$$\vec{\mathbf{a}} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{\mathbf{v}}}{\Delta t} = \frac{d\vec{\mathbf{v}}}{dt}$$



Average and Instantaneous Acceleration

- When a particle's velocity changes in a time interval Δt the average acceleration is:

$$\vec{a}_{avg} \equiv \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_i}{t_f - t_i}$$

- Like previously, the instantaneous acceleration is then:

$$\vec{a} \equiv \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d^2 \vec{r}}{dt^2} = \frac{d}{dt} (v_x \hat{i} + v_y \hat{j} + v_z \hat{k})$$

$$= \frac{dv_x}{dt} \hat{i} + \frac{dv_y}{dt} \hat{j} + \frac{dv_z}{dt} \hat{k}$$

$$= a_x \hat{i} + a_y \hat{j} + a_z \hat{k}$$

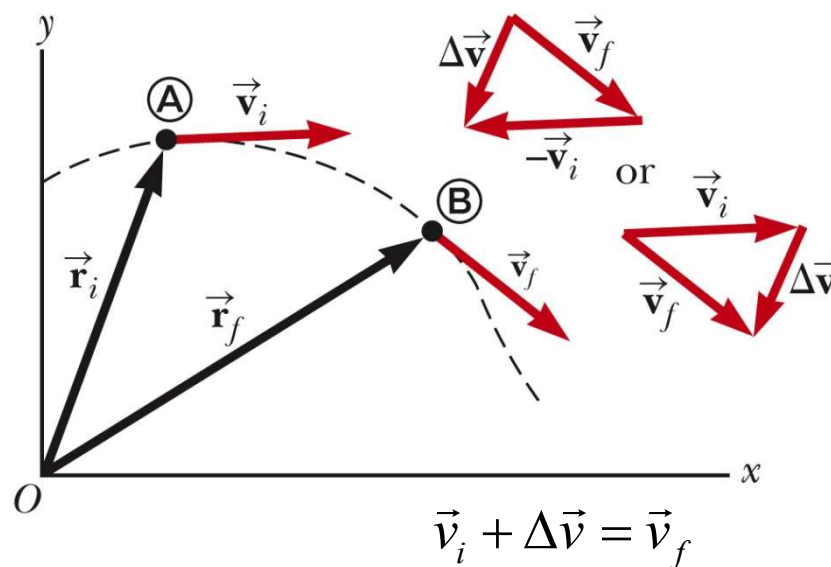
- Scalar components:

$$a_x = \frac{dv_x}{dt}$$

$$a_y = \frac{dv_y}{dt}$$

$$a_z = \frac{dv_z}{dt}$$

$$\vec{v}_f + (-\vec{v}_i) = \Delta \vec{v}$$



Examples

- ▶ Here are four descriptions of the position (in meters) of a puck as it moves in an xy plane:

1. $x = -3t^2 + 4t - 2$ and $y = 6t^2 - 4t$

2. $x = -3t^3 - 4t$ and $y = -5t^2 + 6$

3. $\vec{r} = 2t^2\hat{i} - (4t + 3)\hat{j}$

4. $\vec{r} = (4t^3 - 2t)\hat{i} + 3\hat{j}$

Are the x and y acceleration components constant? Is the acceleration constant?

Examples

- Here are four descriptions of the position (in meters) of a puck as it moves in an xy plane:

1. $x = -3t^2 + 4t - 2$ and $y = 6t^2 - 4t$

$$a_x = \frac{d^2x}{dt^2} = -6 \quad a_y = \frac{d^2y}{dt^2} = 12$$

2. $x = -3t^3 - 4t$ and $y = -5t^2 + 6$

$$a_x = \frac{d^2x}{dt^2} = -18t \quad a_y = \frac{d^2y}{dt^2} = -10$$

3. $\vec{r} = 2t^2\hat{i} - (4t + 3)\hat{j}$

4. $\vec{r} = (4t^3 - 2t)\hat{i} + 3\hat{j}$

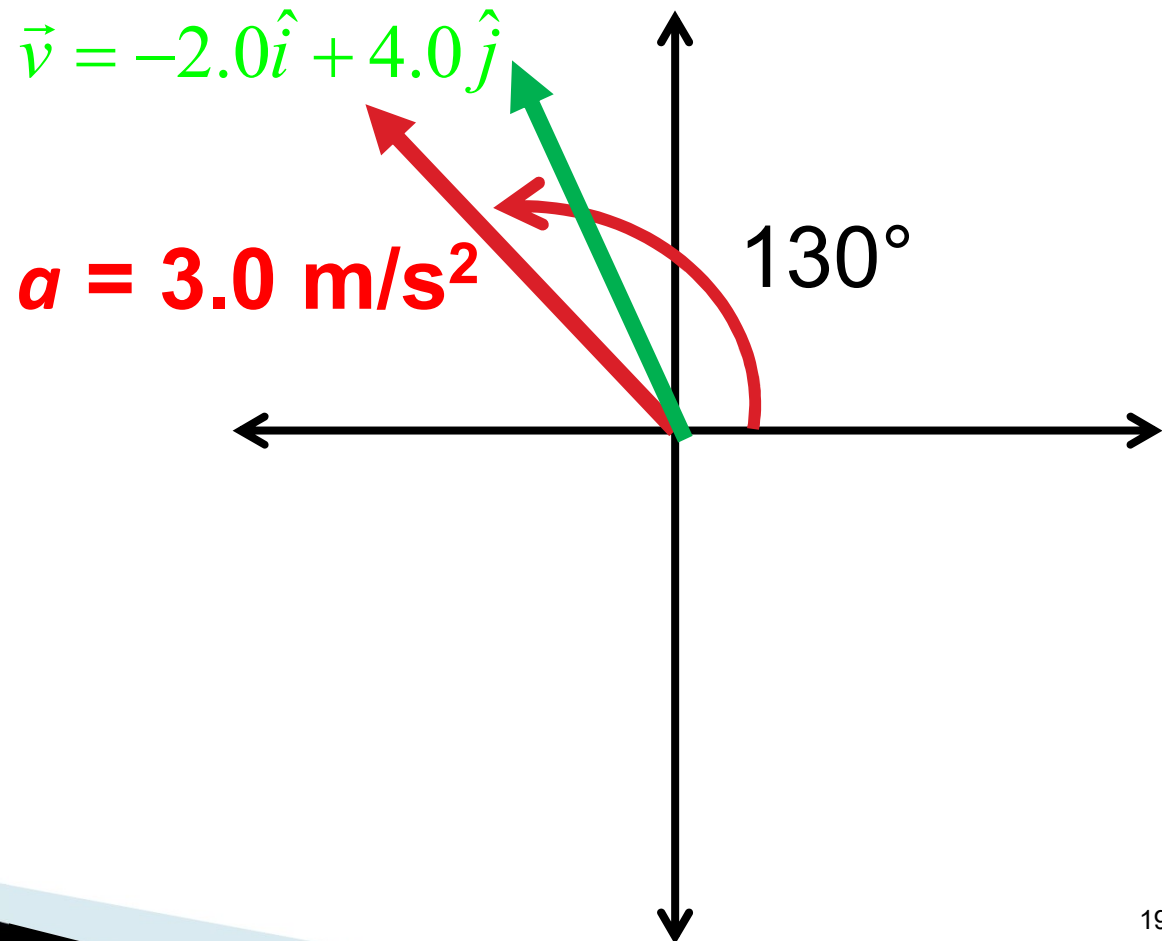
$$a_x = \frac{d^2x}{dt^2} = 4 \quad a_y = \frac{d^2y}{dt^2} = 0$$

Are the x and y acceleration components constant? Is the acceleration constant?

$$a_x = \frac{d^2x}{dt^2} = 24t \quad a_y = \frac{d^2y}{dt^2} = 0$$

Example

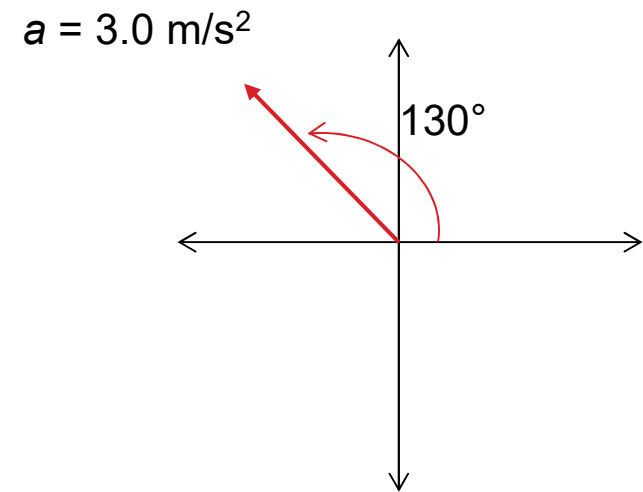
A particle with velocity $\vec{v} = -2.0\hat{i} + 4.0\hat{j}$ (in meters per second) at $t = 0$ undergoes a constant acceleration $\vec{a}$ of magnitude $a = 3.0 \text{ m/s}^2$ at an angle $\theta = 130^\circ$ from the positive direction of the x axis. What is the particle's velocity $\vec{v}$ at $t = 5 \text{ s}$?



Example

A particle with velocity $\vec{v} = -2.0\hat{i} + 4.0\hat{j}$ (in meters per second) at $t = 0$ undergoes a constant acceleration $\vec{a}$ of magnitude $a = 3.0 \text{ m/s}^2$ at an angle $\theta = 130^\circ$ from the positive direction of the x axis. What is the particle's velocity $\vec{v}$ at $t = 5 \text{ s}$?

► What do we know? $a = 3.0 \text{ m/s}^2$ at an angle $\theta = 130^\circ$ to x axis

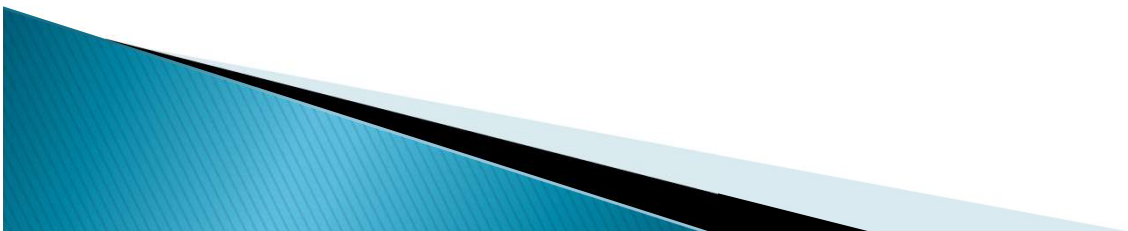


- What magnitude??

- Which angle??

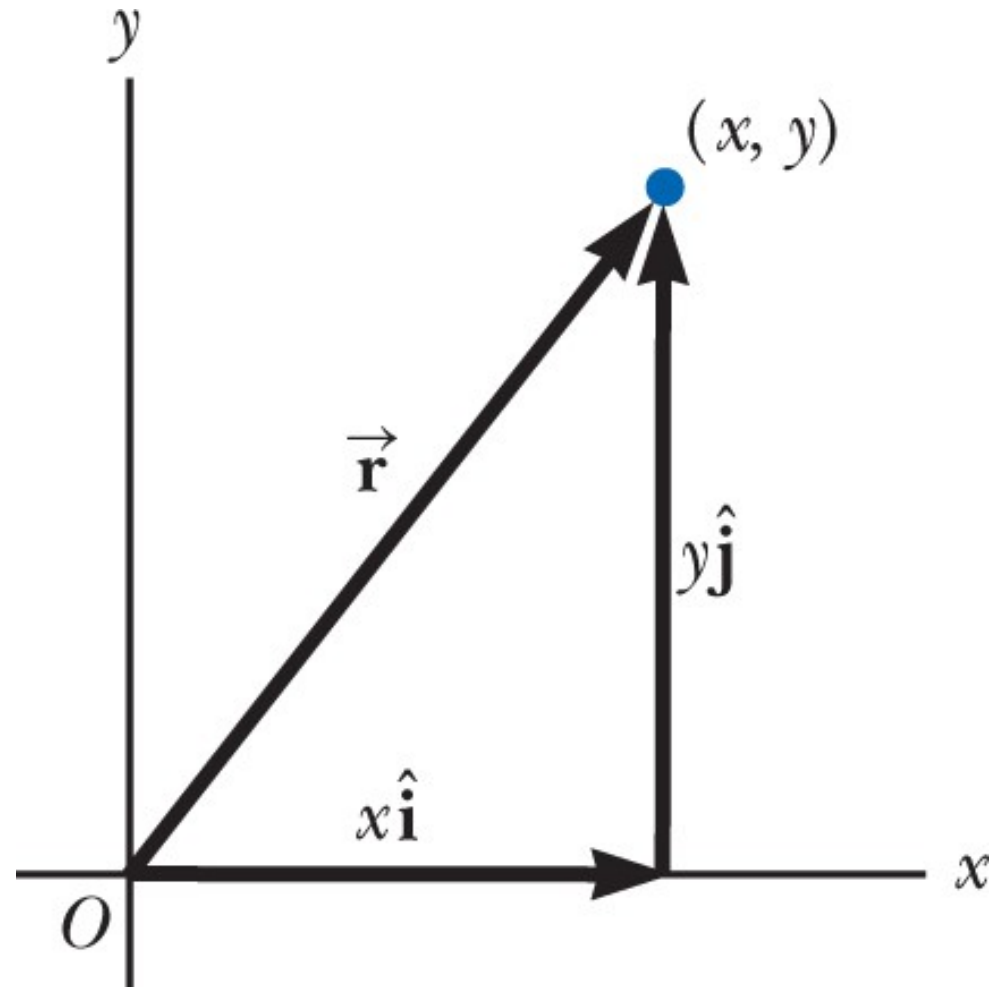
Producing An Acceleration

- ▶ Various changes in a particle's motion may produce an acceleration.
 - The magnitude of the velocity vector may change.
 - The direction of the velocity vector may change.
 - *Even if the magnitude remains constant*
 - Both magnitude and direction may change simultaneously



Two-Dimensional Motion with Constant Acceleration

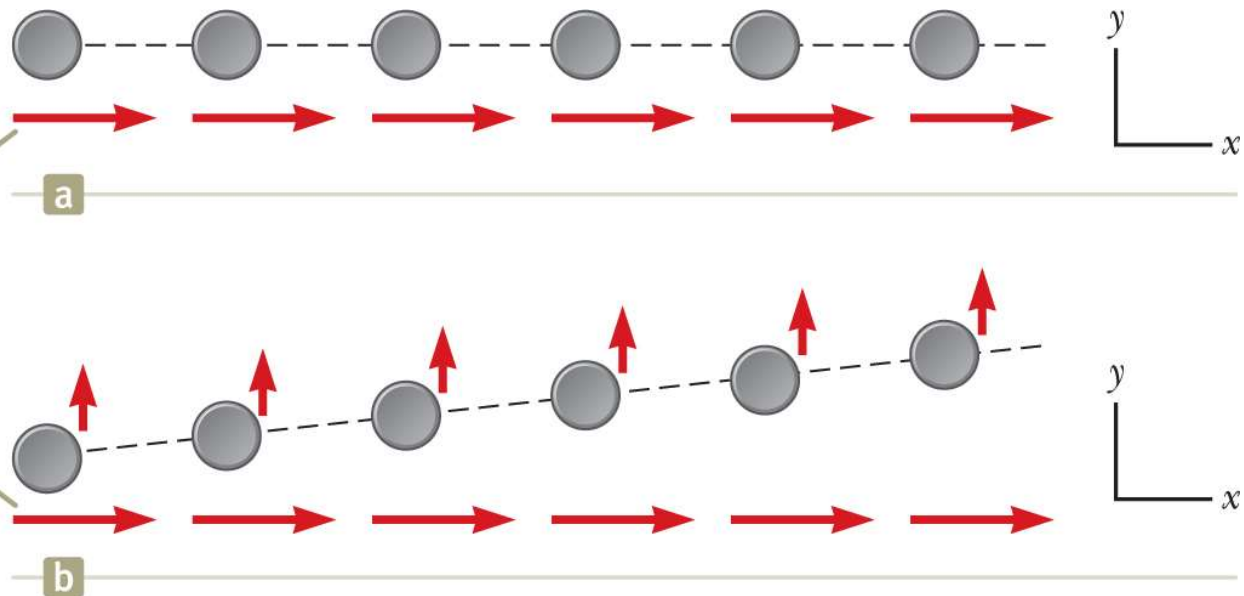
$$\vec{\mathbf{r}} = x\hat{\mathbf{i}} + y\hat{\mathbf{j}}$$



Two-Dimensional Motion with Constant Acceleration

Motion in two dimensions can be modeled as two *independent* motions in each of the two perpendicular directions associated with the x and y axes.

The horizontal red vectors, representing the x component of the velocity, are the same length in both parts of the figure, which demonstrates that motion in two dimensions can be modeled as two independent motions in perpendicular directions.



Two-Dimensional Motion with Constant Acceleration

$$\vec{\mathbf{v}} = \frac{d\vec{\mathbf{r}}}{dt} = \frac{dx}{dt}\hat{\mathbf{i}} + \frac{dy}{dt}\hat{\mathbf{j}} = v_x\hat{\mathbf{i}} + v_y\hat{\mathbf{j}}$$

$$v_{xf} = v_{xi} + a_x t \quad \text{and} \quad v_{yf} = v_{yi} + a_y t$$

$$\vec{\mathbf{v}}_f = (v_{xi} + a_x t)\hat{\mathbf{i}} + (v_{yi} + a_y t)\hat{\mathbf{j}} = (v_{xi}\hat{\mathbf{i}} + v_{yi}\hat{\mathbf{j}}) + (a_x\hat{\mathbf{i}} + a_y\hat{\mathbf{j}})t$$

$$\vec{\mathbf{v}}_f = \vec{\mathbf{v}}_i + \vec{\mathbf{a}}t \quad (\text{for constant } \vec{\mathbf{a}})$$



Two-Dimensional Motion with Constant Acceleration

$$x_f = x_i + v_{xi}t + \frac{1}{2}a_x t^2 \quad y_f = y_i + v_{yi}t + \frac{1}{2}a_y t^2$$

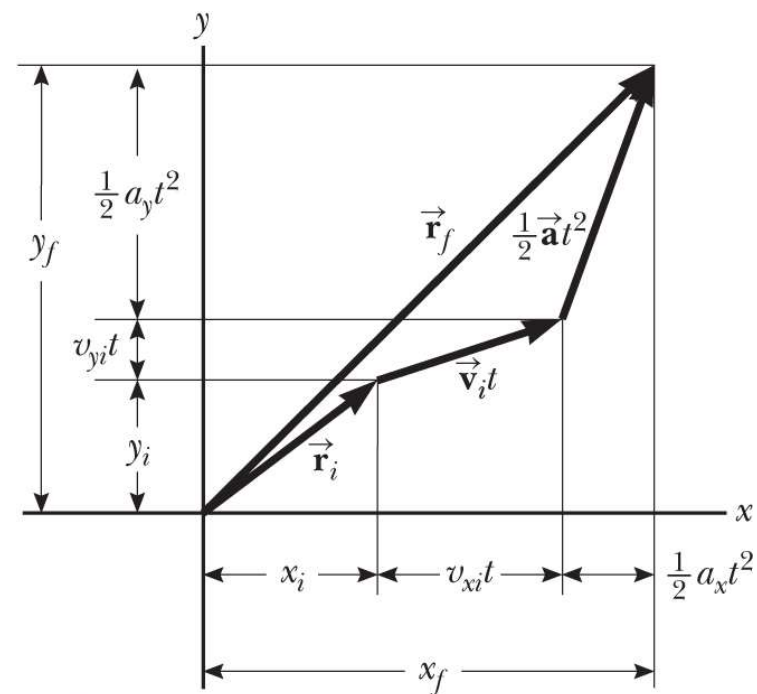
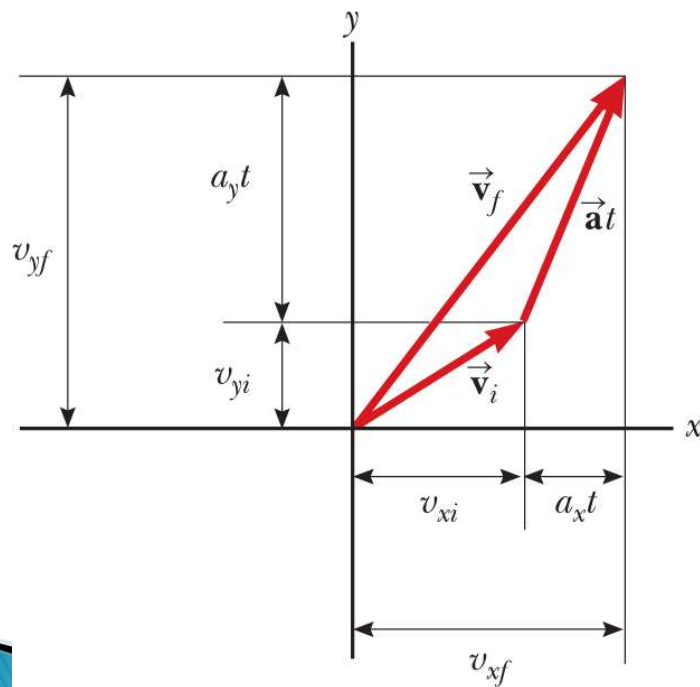
$$\vec{\mathbf{r}}_f = \left(x_i + v_{xi}t + \frac{1}{2}a_x t^2 \right) \hat{\mathbf{i}} + \left(y_i + v_{yi}t + \frac{1}{2}a_y t^2 \right) \hat{\mathbf{j}}$$

$$= \left(x_i \hat{\mathbf{i}} + y_i \hat{\mathbf{j}} \right) + \left(v_{xi} \hat{\mathbf{i}} + v_{yi} \hat{\mathbf{j}} \right) t + \frac{1}{2} \left(a_x \hat{\mathbf{i}} + a_y \hat{\mathbf{j}} \right) t^2$$

$$\vec{\mathbf{r}}_f = \vec{\mathbf{r}}_i + \vec{\mathbf{v}}_i t + \frac{1}{2} \vec{\mathbf{a}} t^2 \quad (\text{for constant } \vec{\mathbf{a}})$$

Two-Dimensional Motion with Constant Acceleration

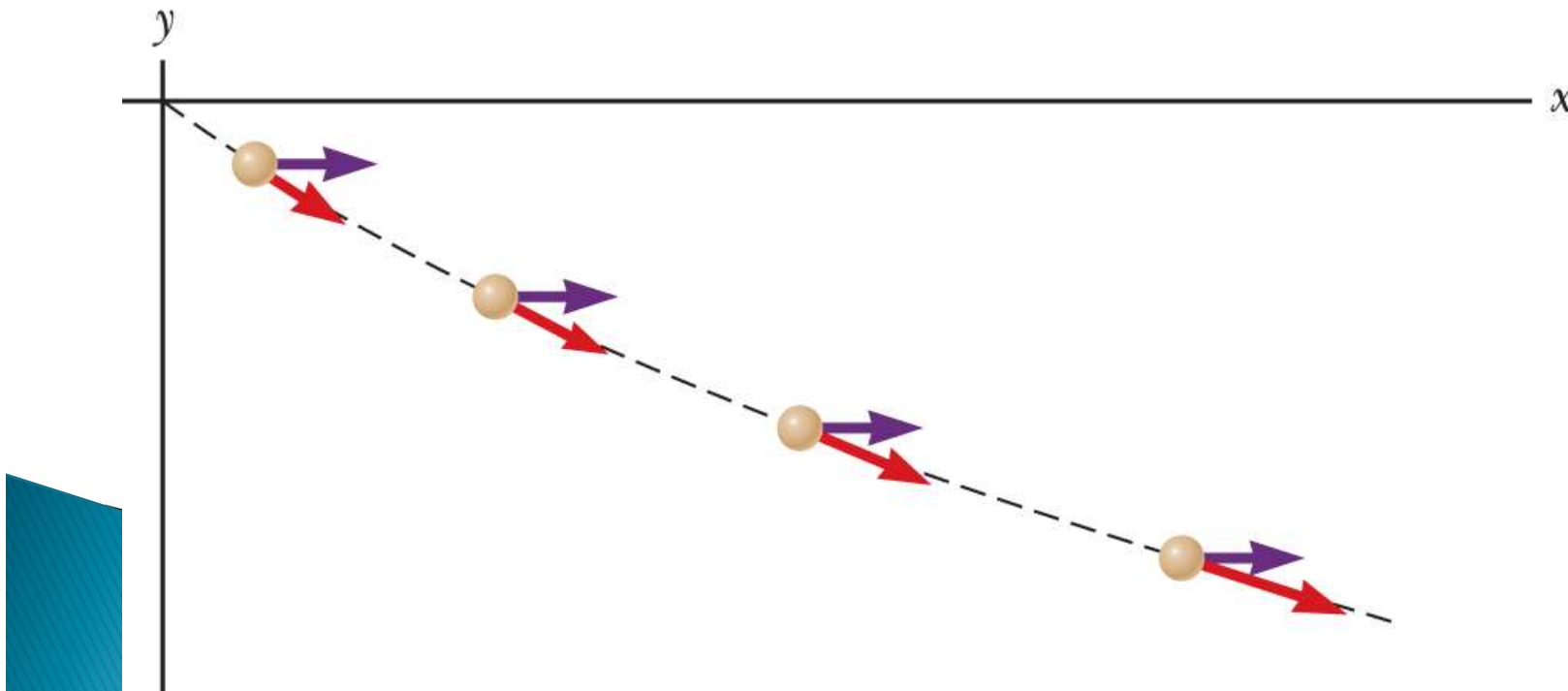
$$\left. \begin{aligned} \vec{\mathbf{v}}_f &= \vec{\mathbf{v}}_i + \vec{\mathbf{a}}t \\ \vec{\mathbf{r}}_f &= \vec{\mathbf{r}}_i + \vec{\mathbf{v}}_i t + \frac{1}{2} \vec{\mathbf{a}}t^2 \end{aligned} \right\} \text{ (for constant } \vec{\mathbf{a}} \text{)}$$



Example 4.1: Motion in a Plane

A particle moves in the xy plane, starting from the origin at $t = 0$ with an initial velocity having an x component of 20 m/s and a y component of -15 m/s. The particle experiences an acceleration in the x direction, given by $a_x = 4.0$ m/s².

(A) Determine the total velocity vector at any later time.



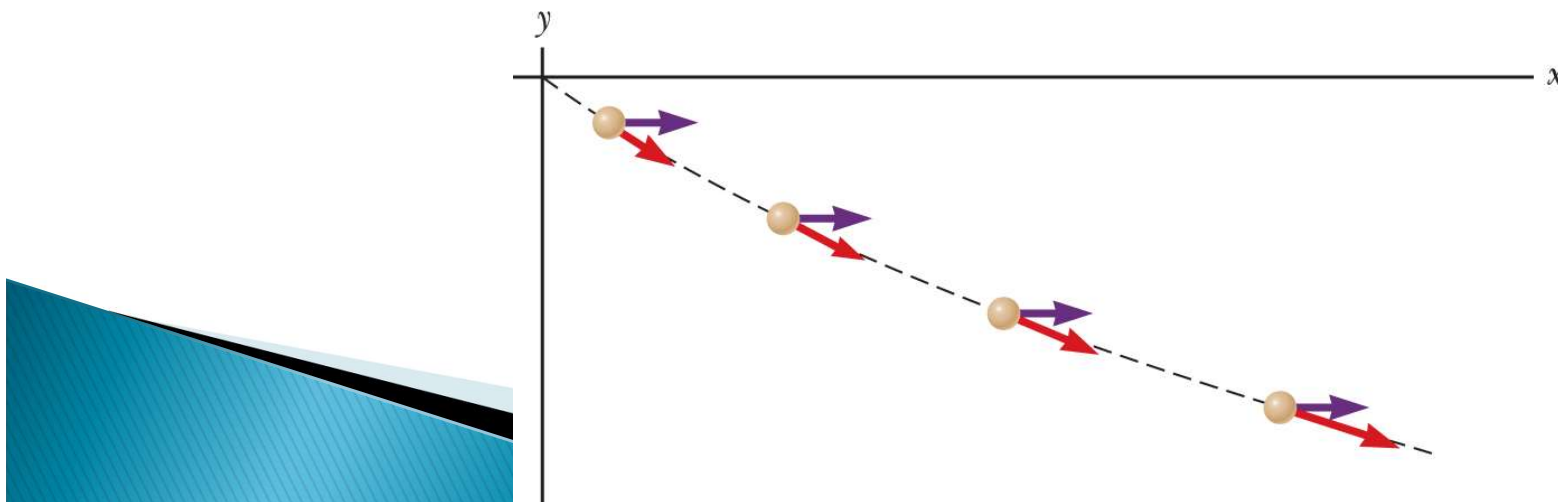
Example 4.1: Motion in a Plane

$$v_{xi} = 20 \text{ m/s}, v_{yi} = -15 \text{ m/s}, a_x = 4.0 \text{ m/s}, a_y = 0$$

$$\vec{\mathbf{v}}_f = \vec{\mathbf{v}}_i + \vec{\mathbf{a}}t = (v_{xi} + a_x t)\hat{\mathbf{i}} + (v_{yi} + a_y t)\hat{\mathbf{j}}$$

$$\vec{\mathbf{v}}_f = [20 + (4.0)t]\hat{\mathbf{i}} + [-15 + (0)t]\hat{\mathbf{j}}$$

$$\vec{\mathbf{v}}_f = \boxed{[20 + (4.0)t]\hat{\mathbf{i}} - 15\hat{\mathbf{j}}}$$



Example 4.1: Motion in a Plane

(B) Calculate the velocity and speed of the particle at $t = 5.0$ s and the angle the velocity vector makes with the x axis.

$$\vec{\mathbf{v}}_f = \left[(20 + 4.0(5.0))\hat{\mathbf{i}} - 15\hat{\mathbf{j}} \right] = \boxed{(40\hat{\mathbf{i}} - 15\hat{\mathbf{j}}) \text{ m/s}}$$

$$\theta = \tan^{-1} \left(\frac{v_{yf}}{v_{xf}} \right) = \tan^{-1} \left(\frac{-15 \text{ m/s}}{40 \text{ m/s}} \right) = -21^\circ$$

$$v_f = |\vec{\mathbf{v}}_f| = \sqrt{v_{xf}^2 + v_{yf}^2} = \sqrt{(40)^2 + (-15)^2} = \boxed{43 \text{ m/s}}$$

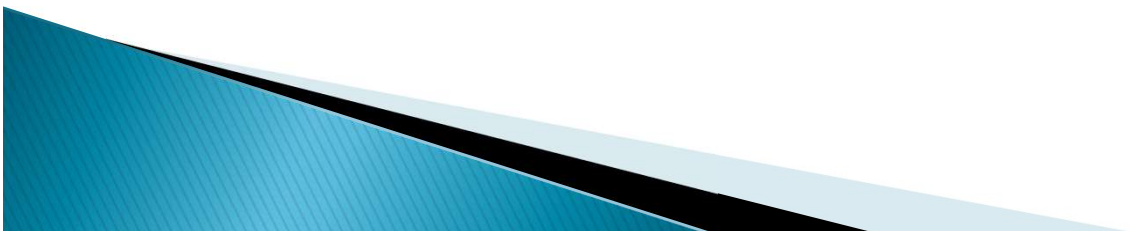

Example 4.1: Motion in a Plane

(C) Determine the x and y coordinates of the particle at any time t and its position vector at this time.

$$\vec{\mathbf{r}}_f = \vec{\mathbf{r}}_i + \vec{\mathbf{v}}_i t + \frac{1}{2} \vec{\mathbf{a}} t^2 = \left(r_{xi} + v_{xi} t + \frac{1}{2} a_x t^2 \right) \hat{\mathbf{i}} + \left(r_{yi} + v_{yi} t + \frac{1}{2} a_y t^2 \right) \hat{\mathbf{j}}$$

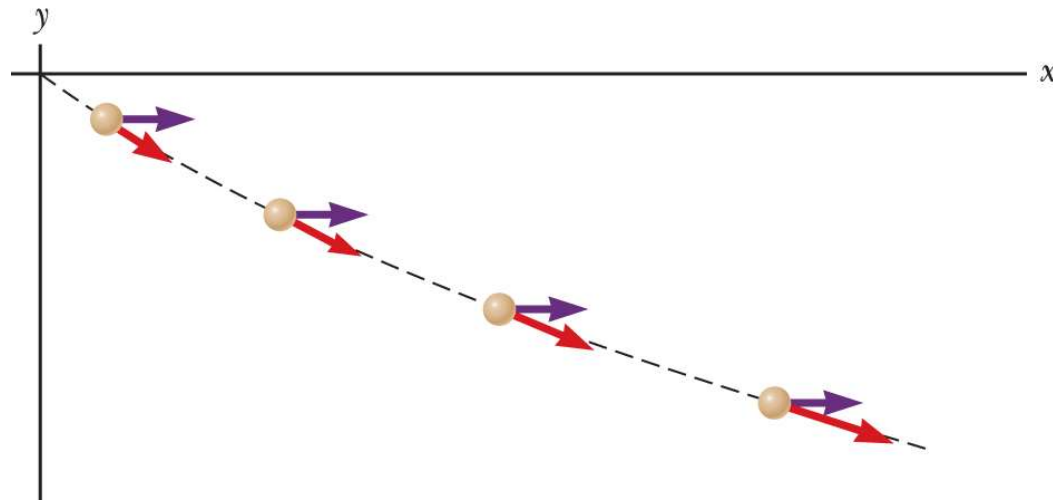
$$\vec{\mathbf{r}}_f = \left[0 + (20)t + \frac{1}{2}(4.0)t^2 \right] \hat{\mathbf{i}} + \left[0 + (-15)t + \frac{1}{2}(0)t^2 \right] \hat{\mathbf{j}}$$

$$\vec{\mathbf{r}}_f = \boxed{(20t + 2.0t^2) \hat{\mathbf{i}} - 15t \hat{\mathbf{j}}}$$



Example 4.1: Motion in a Plane

What if we wait a very long time and then observe the motion of the particle? How would we describe the motion of the particle for large values of the time?

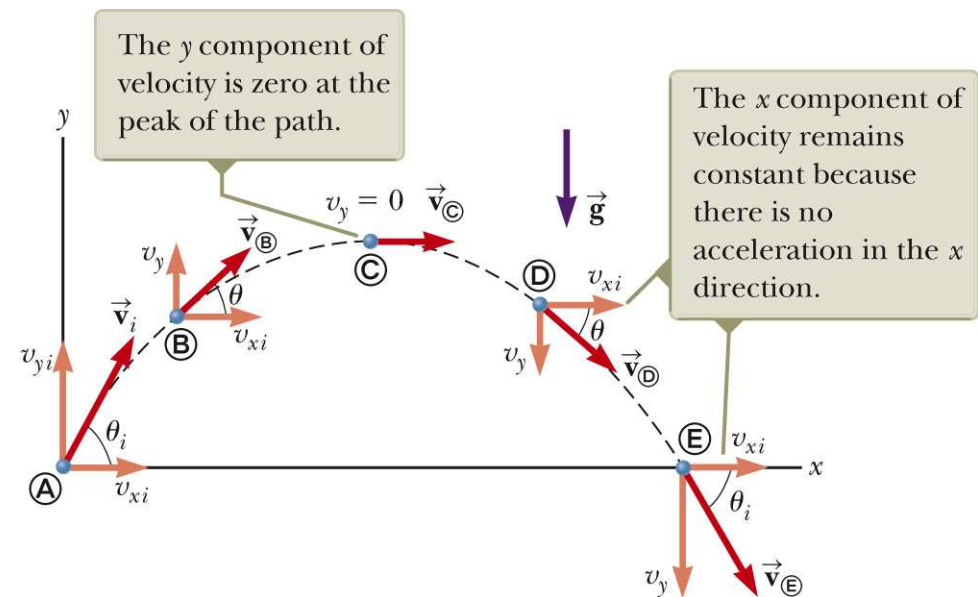
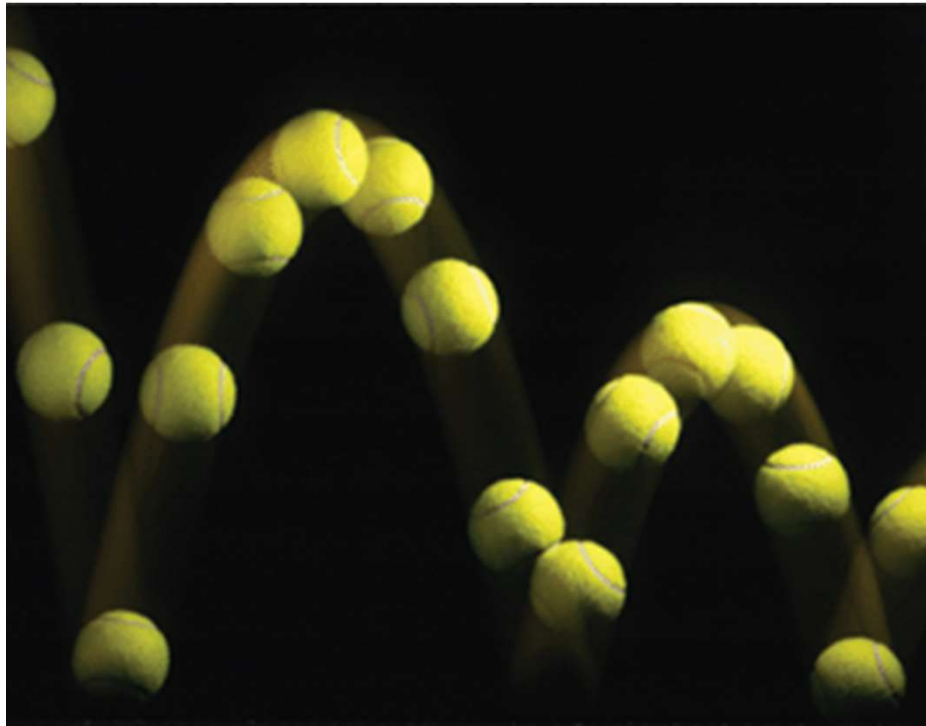


Path becomes more and more parallel to x axis.

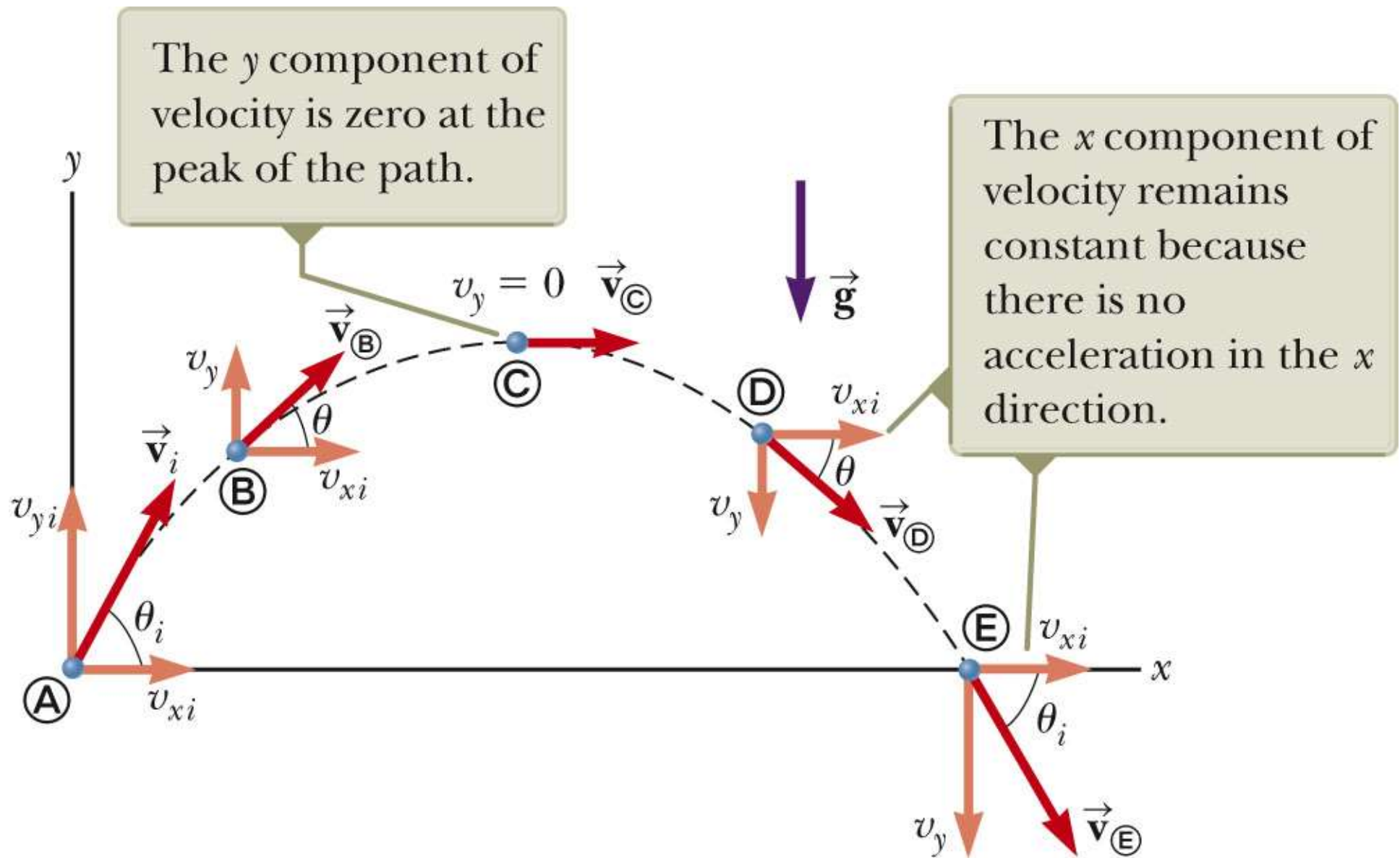
$$\vec{v}_f = [20 + (4.0)t]\hat{i} - 15\hat{j}$$

Projectile Motion

- ▶ Special case: in 2-D motion
- ▶ **Projectile motion**: particle moves in a vertical plane with an initial velocity $\vec{v}_i$ but **acceleration is always the free fall** of gravity downwards $\vec{a} = -\vec{g} = -9.8 \text{ m/s}^2$
- ▶ Assume no air resistance
- ▶ Example of a Path is shown
- ▶ Trajectory is always a parabola



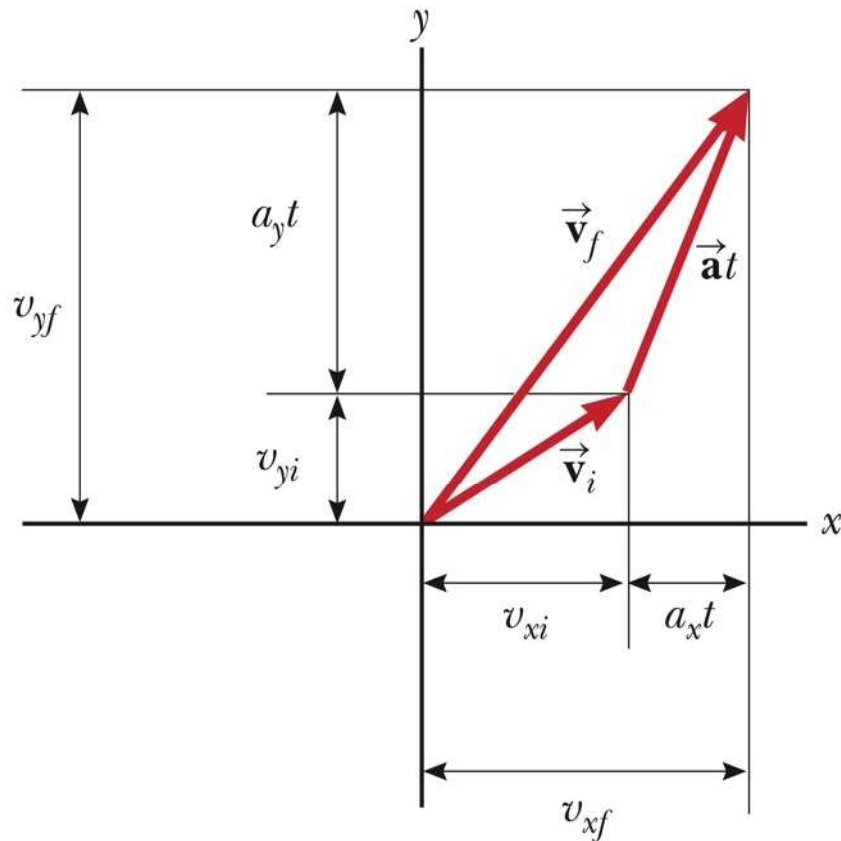
Projectile Motion



Projectile motion

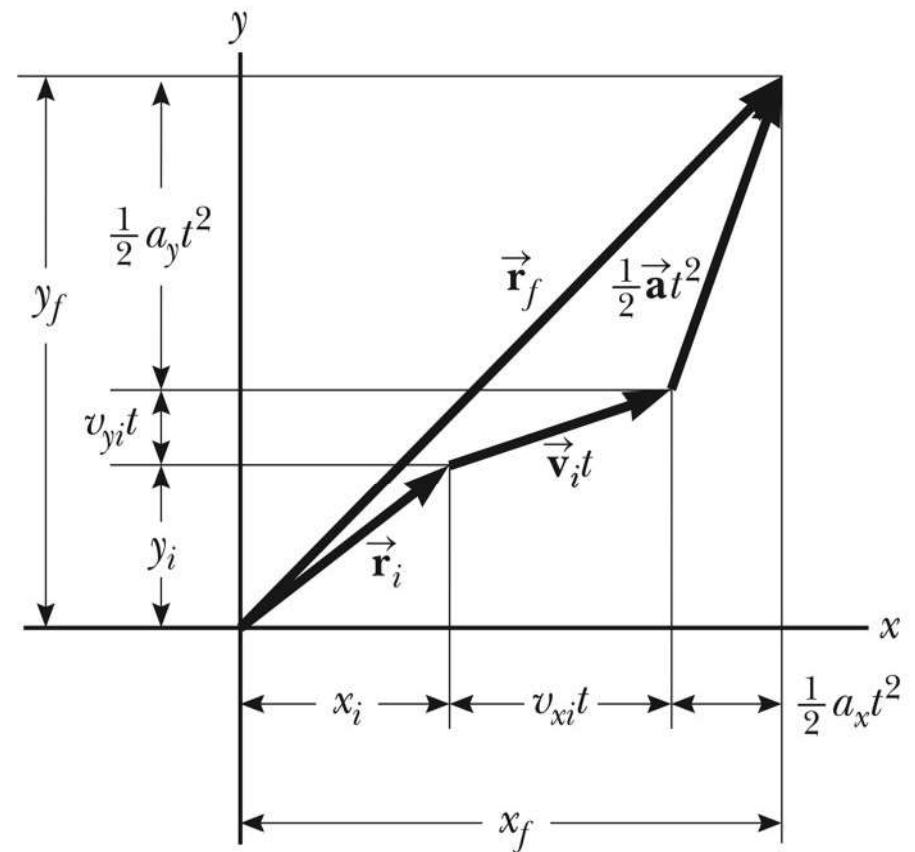
Graphical Representation of the vector equations
Note the Head-to-Tail arrangement!

$$\vec{v}_f = \vec{v}_i + \vec{a}t$$



a

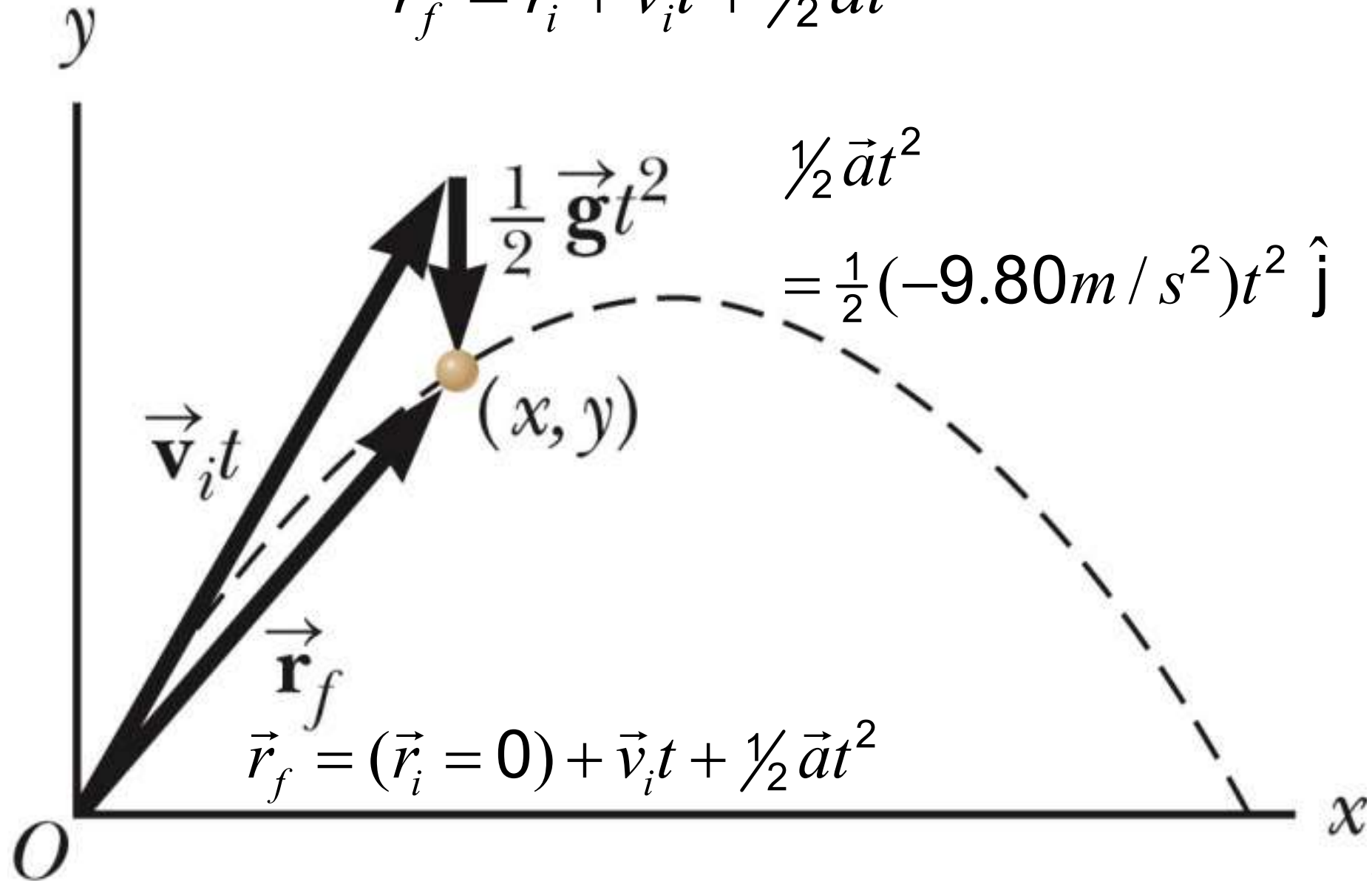
$$\vec{r}_f = \vec{r}_i + \vec{v}_i t + \frac{1}{2} \vec{a} t^2$$



b

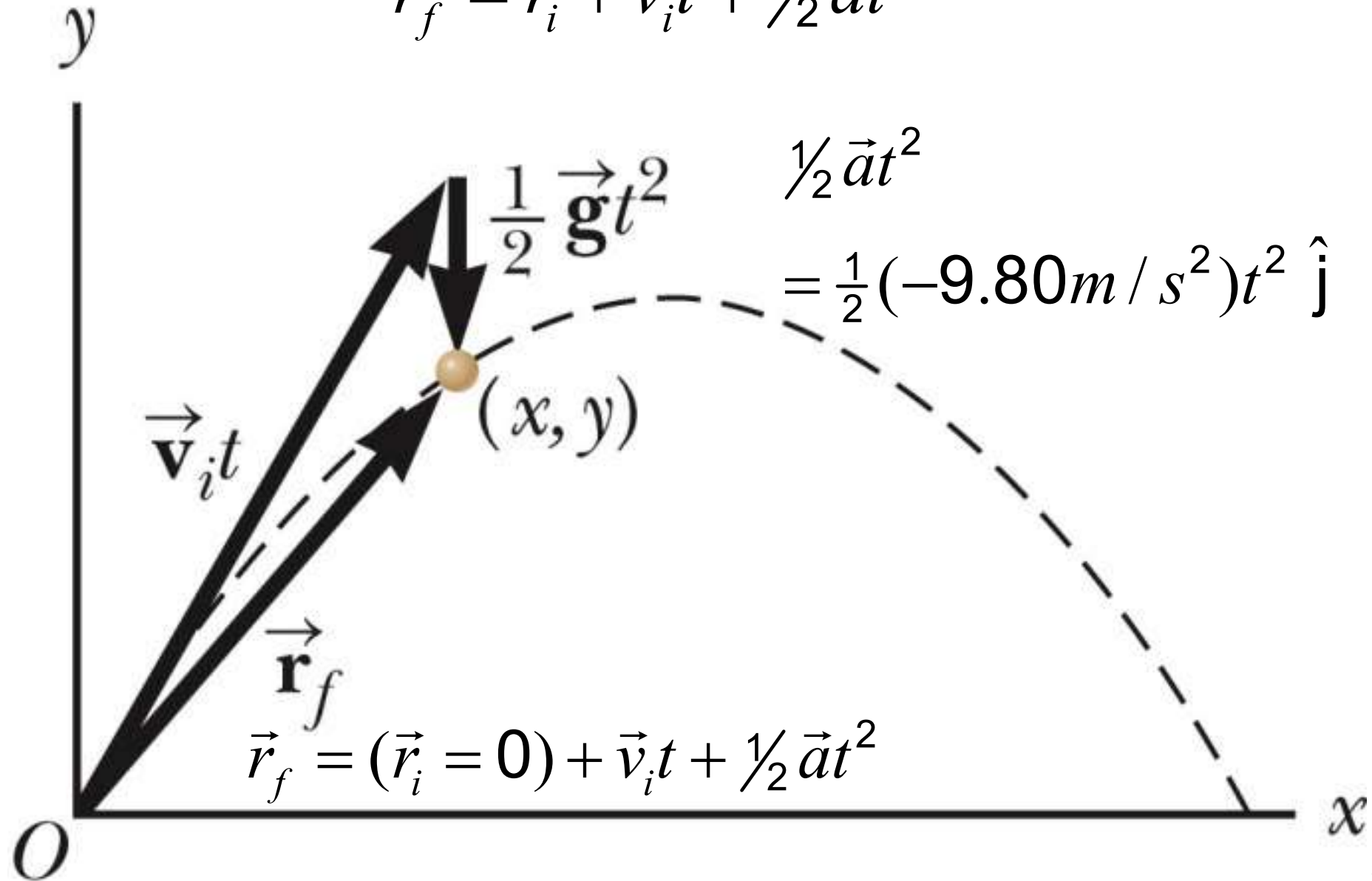
Projectile motion

$$\vec{r}_f = \vec{r}_i + \vec{v}_i t + \frac{1}{2} \vec{a} t^2$$



Projectile motion

$$\vec{r}_f = \vec{r}_i + \vec{v}_i t + \frac{1}{2} \vec{a} t^2$$



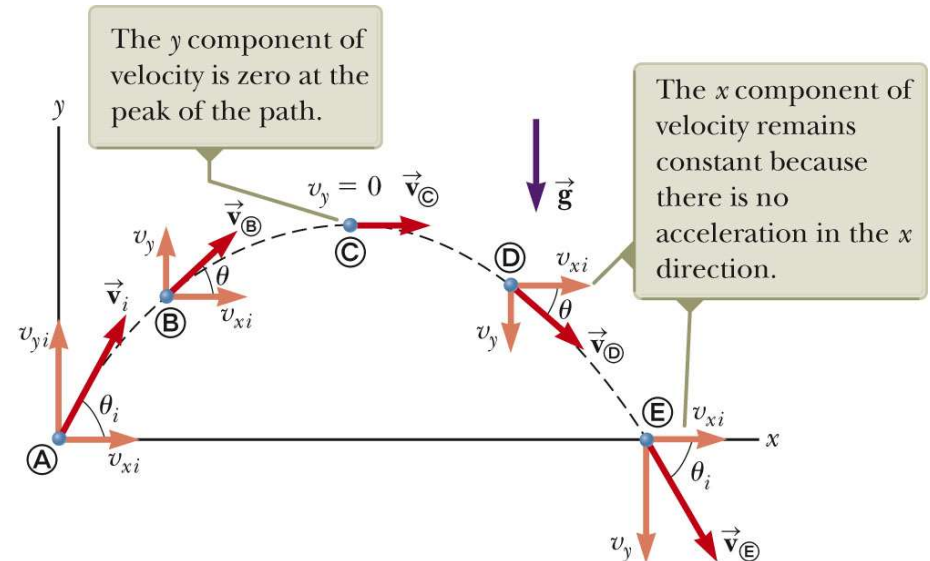
Projectile motion

$$\vec{v}_i = \vec{v}_{xi}\hat{i} + \vec{v}_{yi}\hat{j}$$

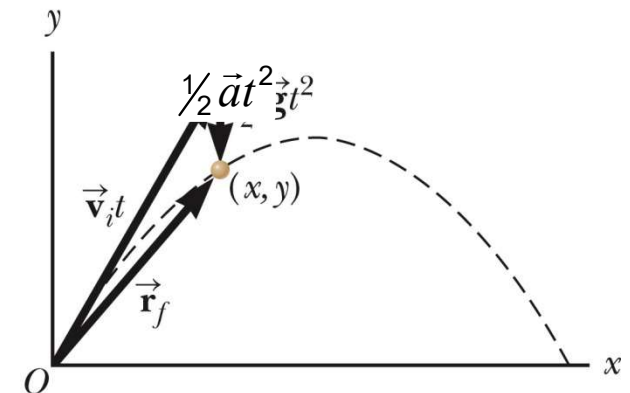
$$v_{xi} = v_i \cos \theta$$

$$v_{yi} = v_i \sin \theta$$

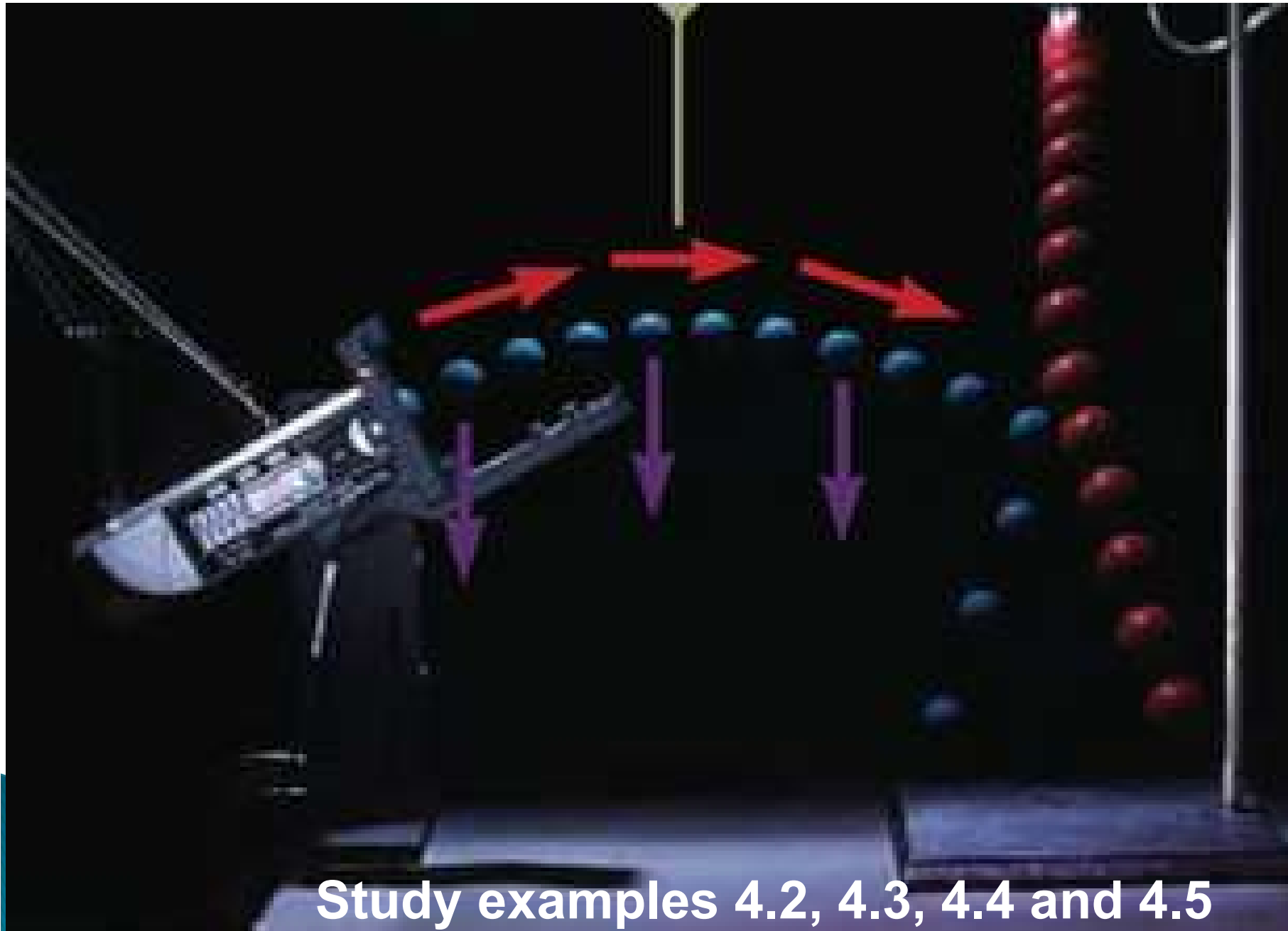
- ▶ Position vector & velocity vector change continuously.
- ▶ **Acceleration is ALWAYS constant**, downwards
- ▶ **NO horizontal acceleration.**
- ▶ The horizontal motion and vertical motion are independent.
- ▶ We can consider the projectile motion as the superposition of motion in the x - and y -directions.
- ▶ Solve the two dimensions separately.



$$\vec{r}_f = \vec{r}_i + \vec{v}_i t + \frac{1}{2} \vec{a} t^2$$



Homework from Serway & Jewett



Study examples 4.2, 4.3, 4.4 and 4.5

Homework from Serway & Jewett

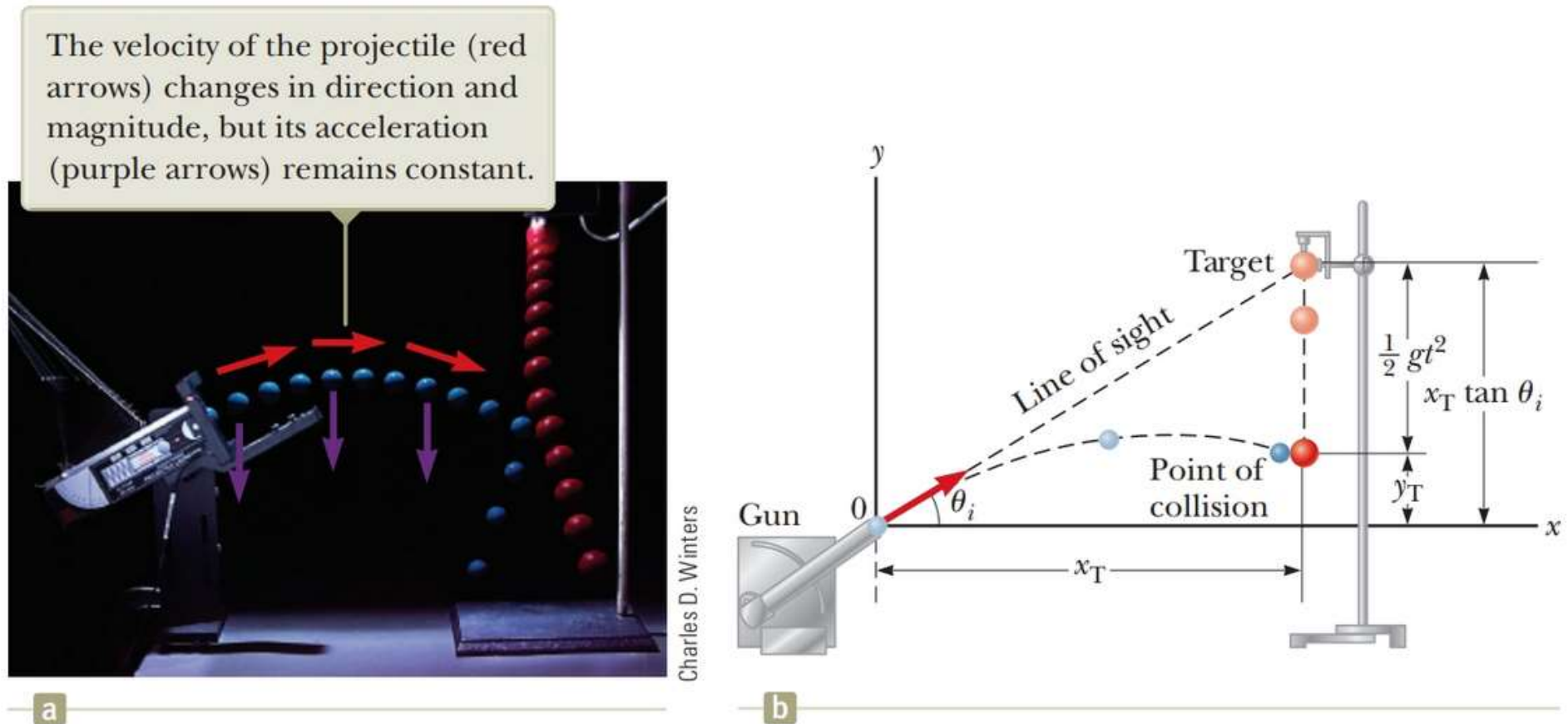


Figure 4.13 (Example 4.3) (a) Multiflash photograph of the projectile–target demonstration. If the gun is aimed directly at the target and is fired at the same instant the target begins to fall, the projectile will hit the target. (b) Schematic diagram of the projectile–target demonstration.

Study examples 4.2, 4.3, 4.4 and 4.5

Projectile Motion Analysed

▶ **Maximum height: h**

▶ **Range: R**

▶ Time to reach h

$$v_{yf} = v_{yi} - gt$$

$$0 = v_i \sin \theta_i - gt$$

$$t = \frac{v_i \sin \theta_i}{g}$$

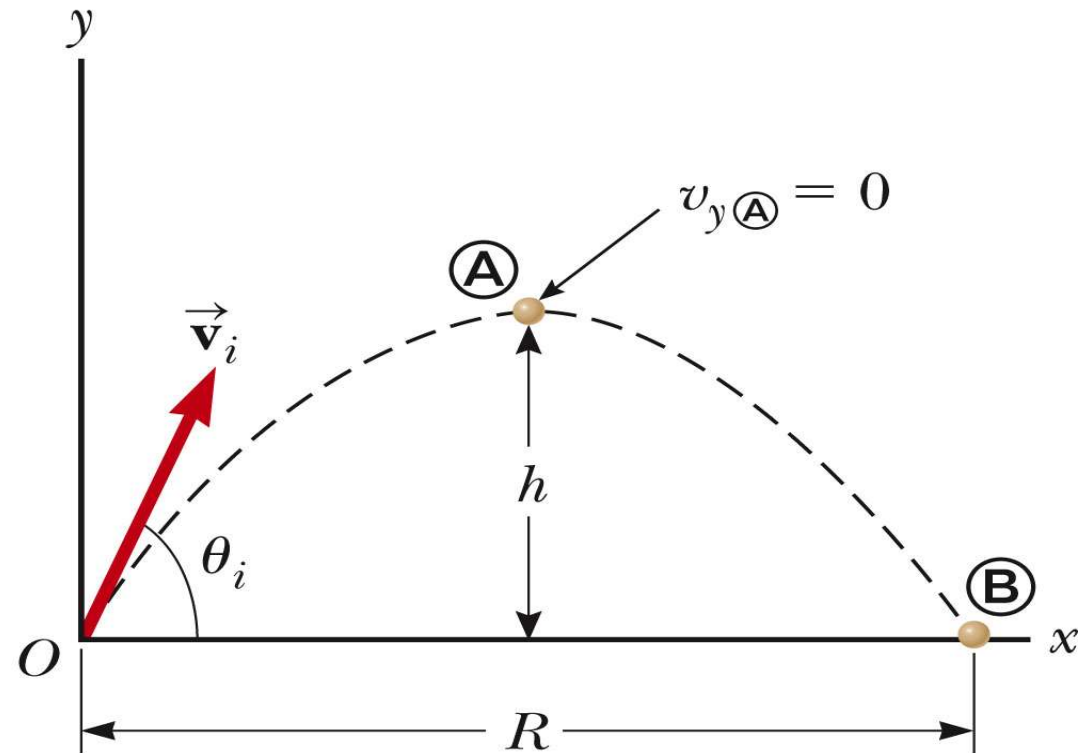
▶ Substituting this t into

$$y_f = y_i + v_{yi}t + \frac{1}{2}a_y t^2$$

$$h = 0 + v_i \sin \theta_i \left(\frac{v_i \sin \theta_i}{g} \right) - \frac{1}{2}g \left(\frac{v_i \sin \theta_i}{g} \right)^2$$

$$h = \frac{v_i^2 \sin^2 \theta_i}{2g}$$

A Required Derivation!



Projectile Motion Analysed

▶ Time of flight: $t_R = 2t_h = \frac{2v_i \sin \theta_i}{g}$

TOF = 2 x the time to reach maximum height

▶ Determining the Range, R :

▶ Horizontal velocity is constant

$$x_{xf} = x_{xi} + v_i t$$

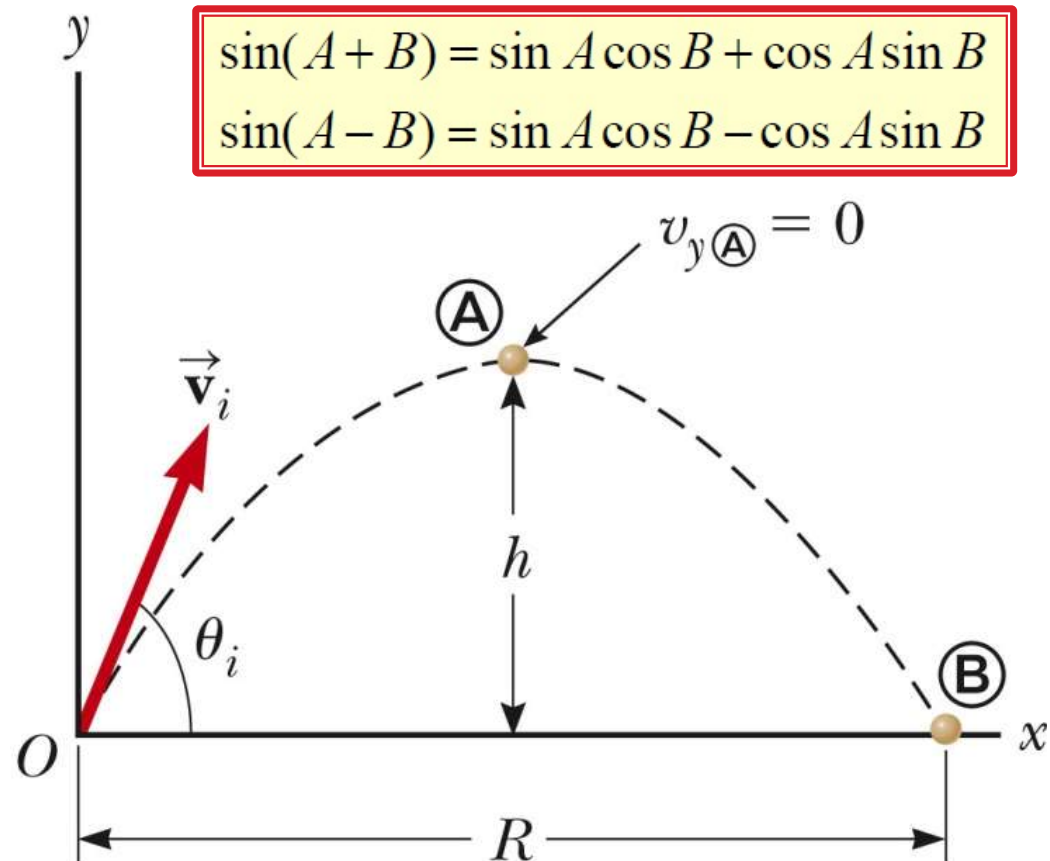
$$R = 0 + (v_i \cos \theta_i) \left(\frac{2v_i \sin \theta_i}{g} \right)$$

$$= \left(\frac{2v_i^2 \sin \theta_i \cos \theta_i}{g} \right)$$

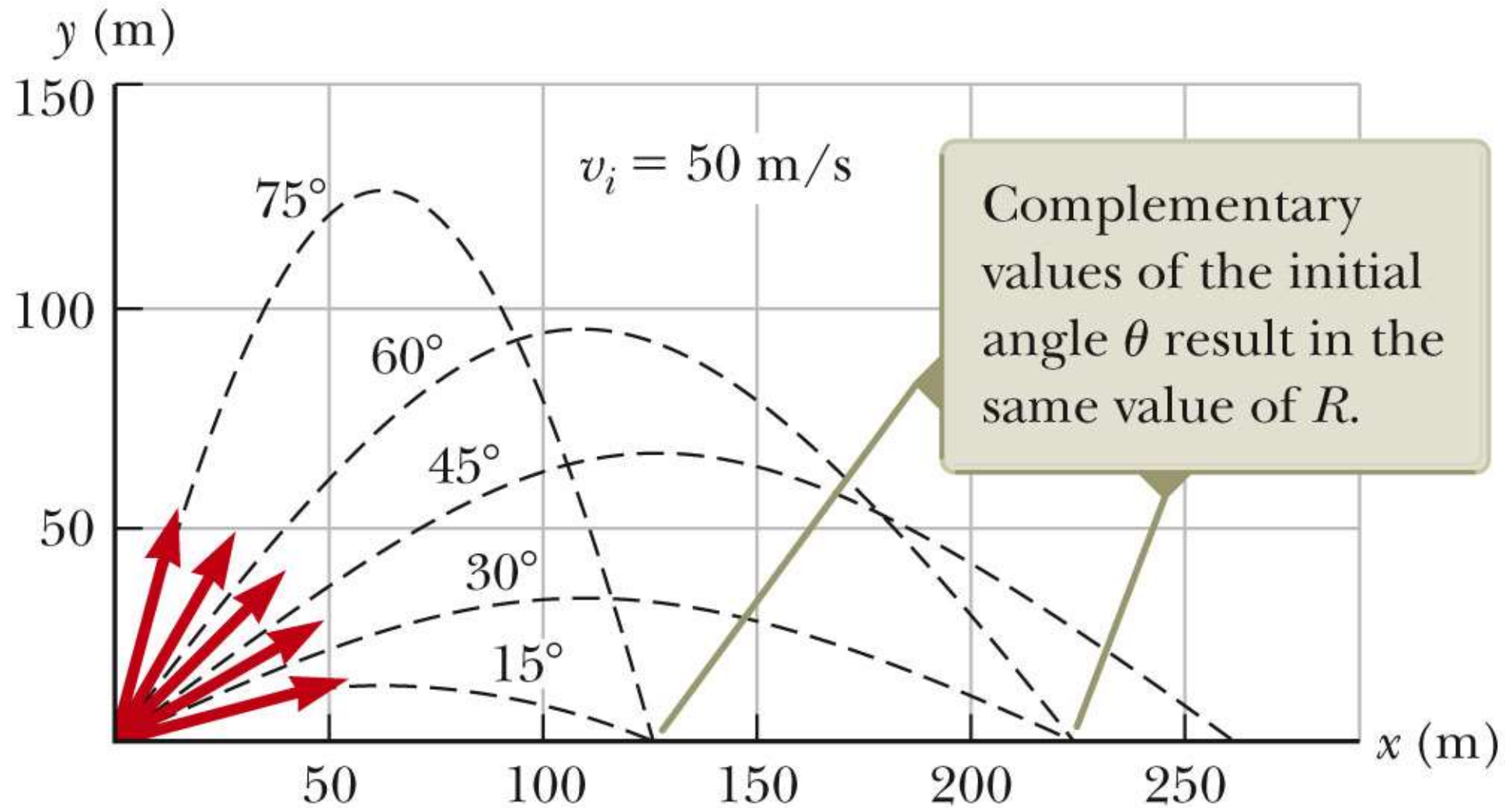
▶ Max R when $\theta = 45^\circ$

A Required Derivation!

$$R = \frac{v_i^2}{g} \sin 2\theta_i$$

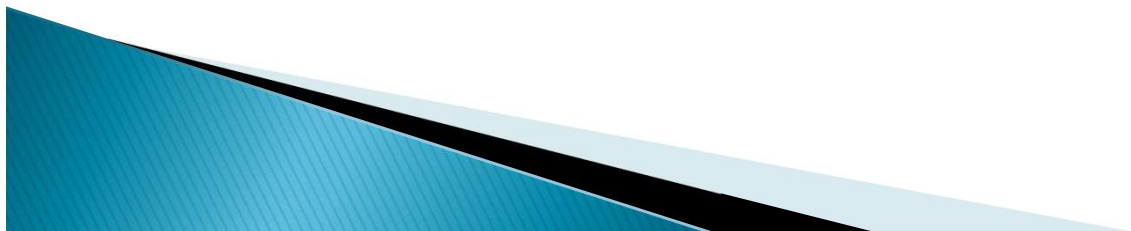


Projectile Motion Analysed



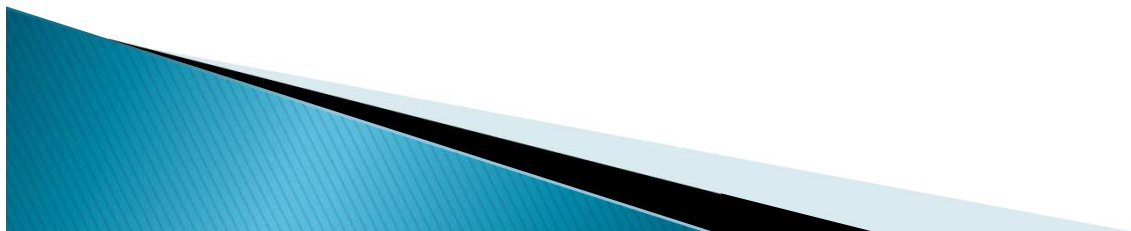
Problem solving strategy

- ▶ Conceptualize
- ▶ Categorize – Choose axes (x – y), free fall (*vertical*)? constant velocity (*horizontal*)
- ▶ Analyse – Separate x and y variables and find components of variables
- ▶ Finalize – Check to see if your answers are consistent with what should happen – are they realistic?



Problem

- ▶ A rifle with a muzzle velocity $v_i = 450$ m/s shoots a bullet at a target 50 m away. How high above the target must the barrel be pointed? Assume the muzzle and the target are at the same height.



Problem

A rifle with a muzzle velocity $v_i = 450 \text{ m/s}$ shoots a bullet at a target 50 m away.

(a) How high above the target must the barrel be pointed?

Assume the muzzle and the target are at the same height.

(b) Determine the maximum height of the trajectory.

► Given:

- $v_i = 450 \text{ m/s}$

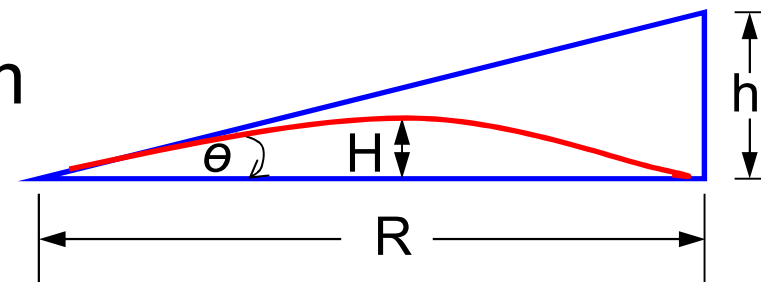
- $R = 50 \text{ m}$

- $g = 9.8 \text{ m/s}^2$

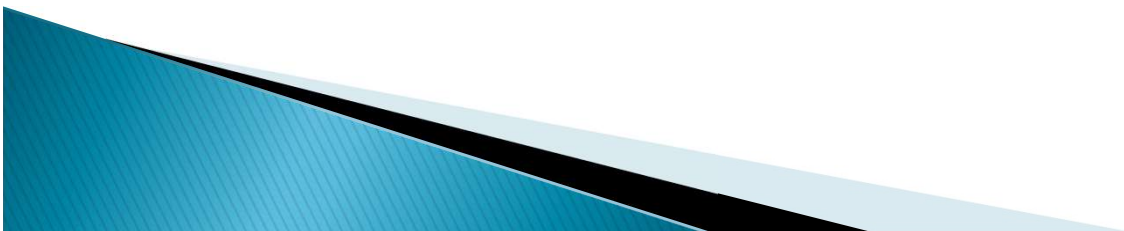
Height aimed at, h

$$\tan \theta = \frac{h}{50}$$

$$h = 60 \text{ mm}$$



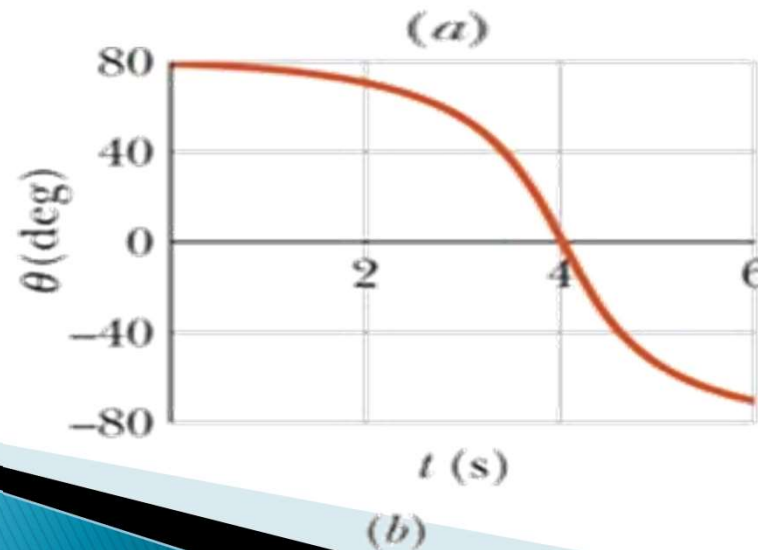
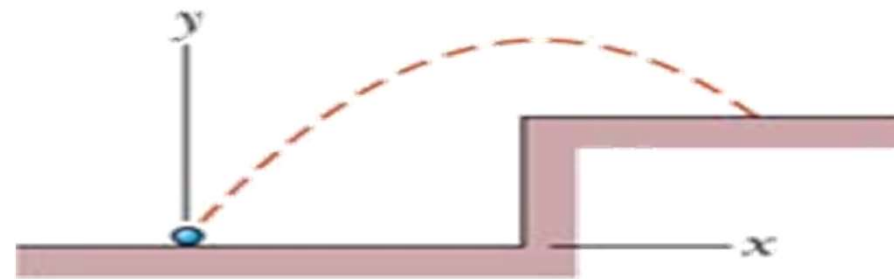
Maximum Height, H



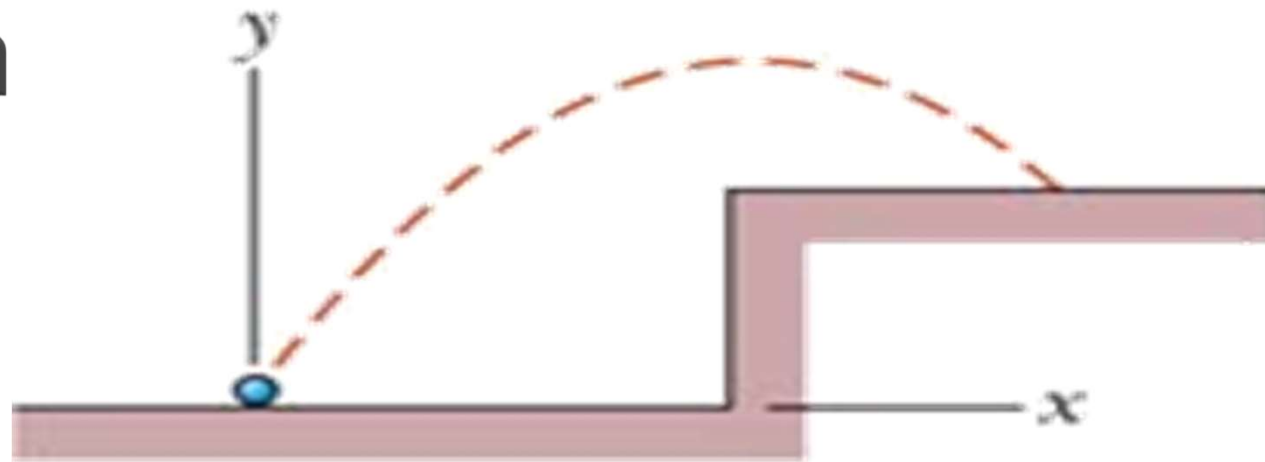
Problem

At time $t = 0$ s, a golf ball is shot from ground level into the air, as shown in the figure. The angle θ between the ball's direction of travel and the positive x axis is given as a function of time t . The ball lands at $t = 6.00$ s.

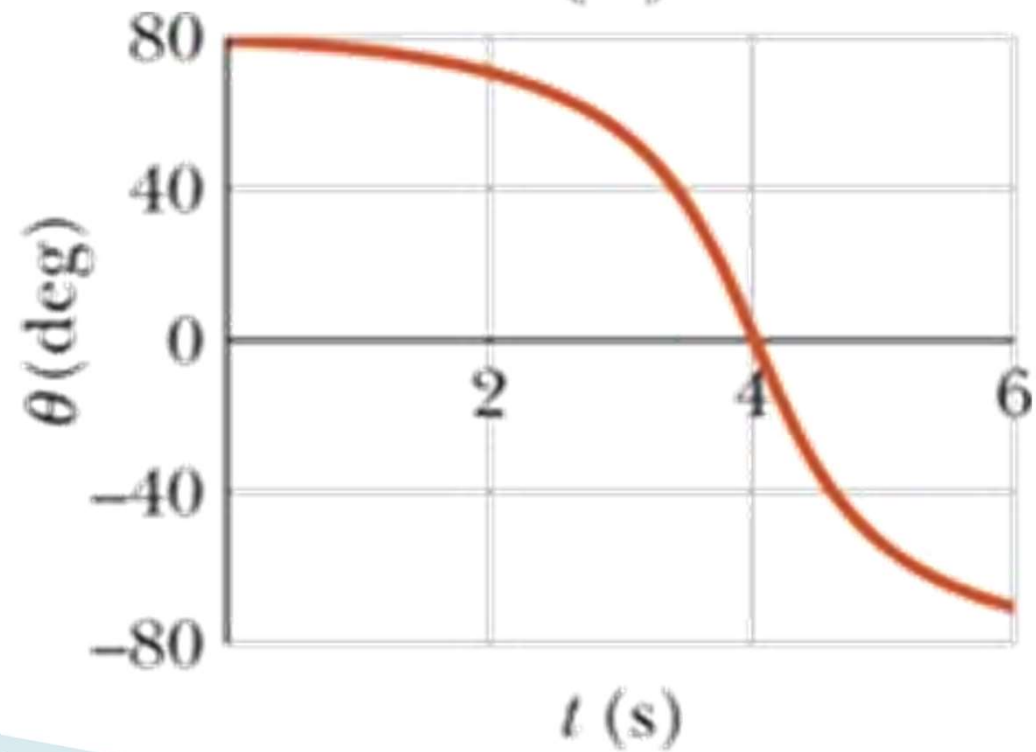
- (a) What is the magnitude v_0 of the ball's launch velocity?
- (b) At what height (y_f) above the launch level does the ball land?
- (c) What is the ball's direction of travel just before it lands?



Problem



(a)



(b)

Problem

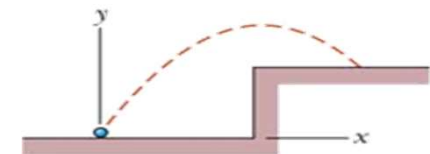
- ▶ At time $t = 0$ s, a golf ball is shot from ground level into the air, as shown in the figure. The angle θ between the ball's direction of travel and the positive x axis is given as a function of time t . The ball lands at $t = 6.00$ s.
 - (a) What is the magnitude v_0 of the ball's launch velocity?
 - (b) At what height (y_f) above the launch level does the ball land?
 - (c) What is the ball's direction of travel just before it lands?

$$v_y = v_i \sin \theta - gt \quad \text{Max height @ 4 s}$$

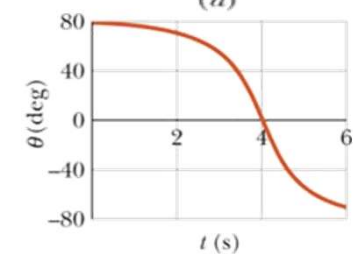
$$0 = v_i \sin(80^\circ) - (9.8)(4)$$

$$v_i = 39.8 \text{ m/s}$$

Ball lands @ 6 s

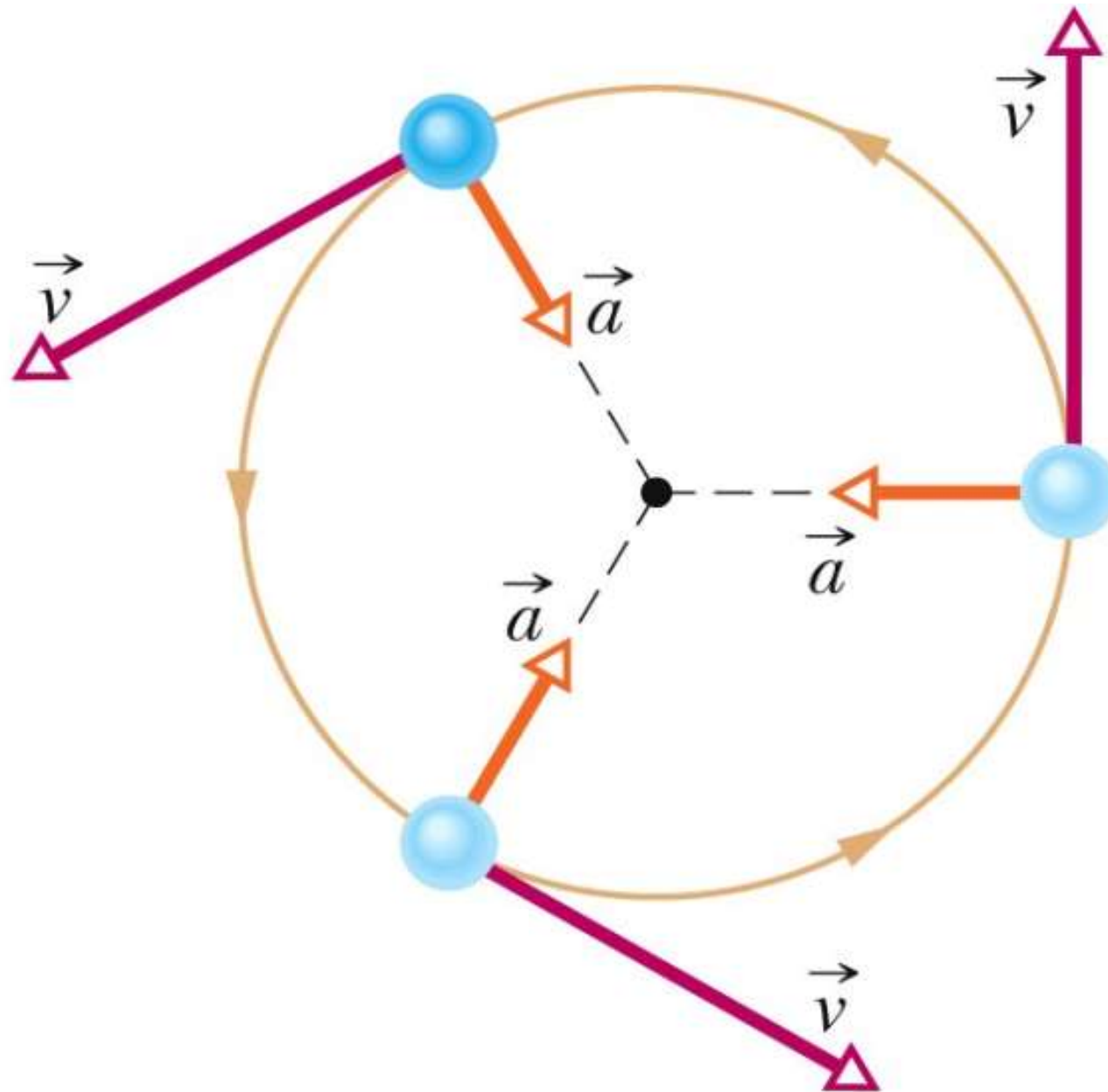


(a)



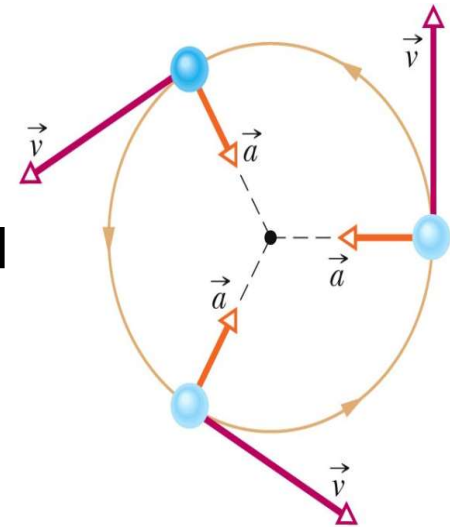
(b)

Uniform Circular Motion



Uniform Circular Motion

- ▶ A particle is in **uniform circular motion** if:
 - It travels in a circular arc
 - At a constant speed
- ▶ Although speed is constant, there is acceleration
 - Velocity is **changing direction**
 - Velocity is directed **tangentially** to the circle
 - Acceleration is directed to centre of the circle – **centripetal acceleration**.
 - If there was an acceleration component parallel to the velocity vector, the particle's speed would increase or decrease = change!
⇒ no longer uniform circular motion



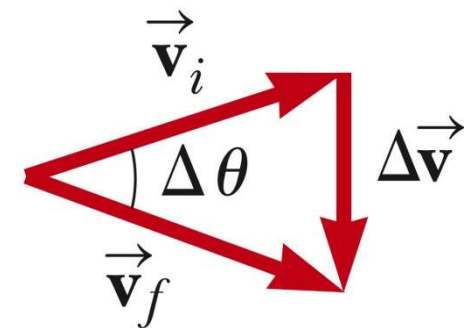
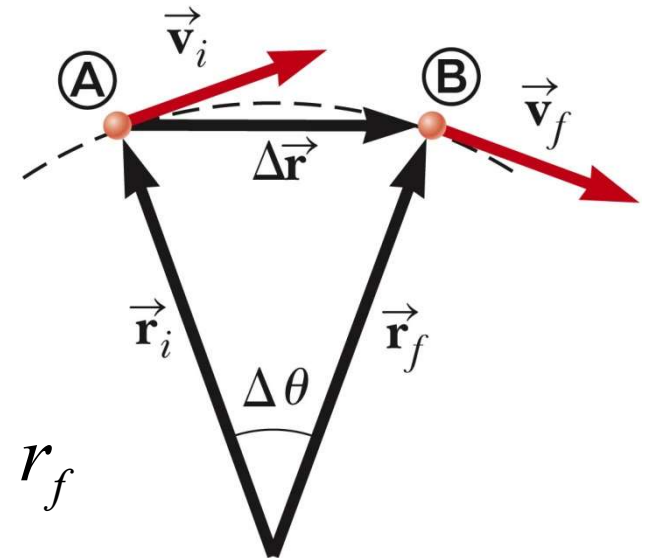
Uniform Circular Motion

- ▶ Consider the diagram with a particle moving from A to B in a circular arc:

- Note: Position vectors, displacement, Δt
- Velocity vectors at A and B, and $\Delta \vec{v}$
- Positions triangle *similar* to velocities triangle.
- $\Delta \theta$ between the position vectors = $\Delta \theta$ between the velocity vector. **Similar Δ 's.**
- $\frac{|\Delta \vec{v}|}{v} = \frac{|\Delta \vec{r}|}{r}$, where $v = v_i = v_f$ and $r = r_i = r_f$

$$\therefore \left| \vec{a}_{avg} \right| = \frac{\frac{|\Delta \vec{v}|}{v}}{\frac{|\Delta t|}{r}} = \frac{v |\Delta \vec{r}|}{r \Delta t} \quad \text{since} \quad |\Delta \vec{v}| = \frac{v}{r} |\Delta \vec{r}|$$

In uniform circular motion, the magnitudes of both position vectors and velocity vectors indexed respectively by i & f , are equal.



Red & black Δ 's are similar!

Uniform Circular Motion

$$|\vec{a}_{avg}| = \frac{|\Delta \vec{v}|}{|\Delta t|} = \frac{v |\Delta \vec{r}|}{r \Delta t} \quad \text{since} \quad |\Delta \vec{v}| = \frac{v |\Delta \vec{r}|}{r}$$

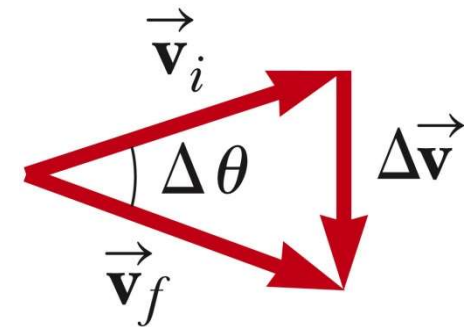
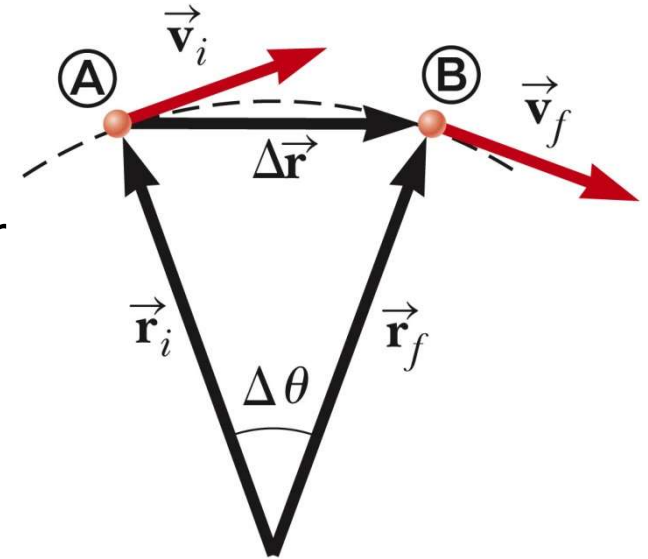
- As $\Delta t \rightarrow 0$, $|\Delta \vec{r}| \rightarrow d\vec{r}$ distance on circular path.

$$\frac{|\Delta \vec{r}|}{\Delta t} \rightarrow v \quad \text{as} \quad \Delta t \rightarrow 0$$

$$\therefore \vec{a}_c = -\frac{v^2}{r} \hat{r} \quad \text{or wrt magnitude only:} \quad a_c = \frac{v^2}{r}$$

- centripetal acceleration (centre-seeking)

A Required Derivation!



Uniform Circular Motion

- Period (T) = time to complete one revolution

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

$$\therefore T = \frac{\text{distance}}{\text{speed}} = \frac{2\pi r}{v}$$

- Angular speed $\omega = \#$ **revolutions per second**
- SI units are Radians/s (Lookup how to convert from $^\circ$ to radians)

$$\omega = \frac{2\pi}{T} \quad \text{2}\pi \text{ radians covered in 1 revolution!}$$

- Combining these two equations:

$$\omega = 2\pi \left(\frac{v}{2\pi r} \right) = \frac{v}{r}$$

$$\therefore v = r\omega$$

- Substituting into acceleration:

$$a_c = \frac{v^2}{r} = \frac{(r\omega)^2}{r} = r\omega^2$$

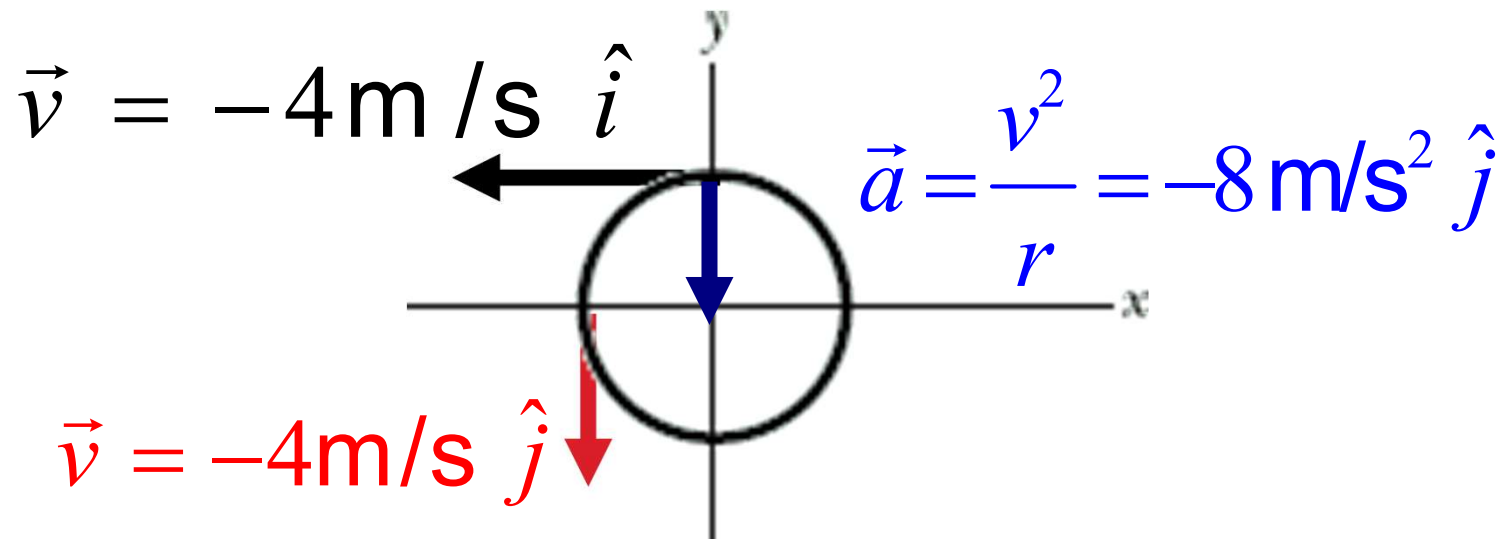
Example

- ▶ An object moves at constant speed along a circular path in a horizontal xy plane, with the centre at the origin. When the object is at $x = -2$ m, its velocity is $-(4\text{m/s})\hat{j}$. Give the object's
 - (a) velocity and
 - (b) acceleration at $y = 2$ m.



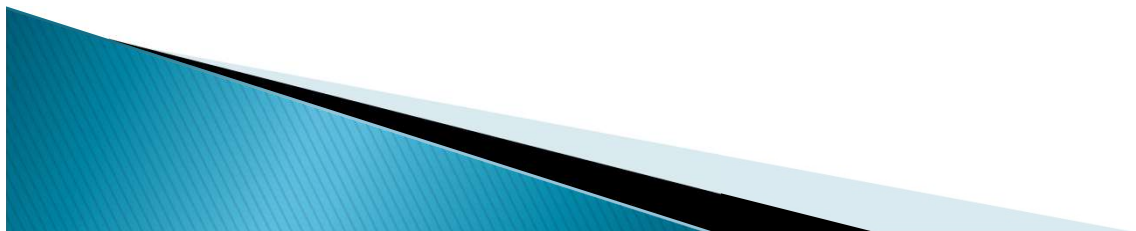
Example

- ▶ An object moves at constant speed along a circular path in a horizontal xy plane, with the centre at the origin. When the object is at $x = -2\text{m}$, its **velocity is** $-(4\text{m/s})\hat{j}$. Give the object's (a) **velocity** and (b) **acceleration** at $y = 2\text{ m}$.



Problem

- ▶ A satellite is at an altitude $h = 200$ km above the earth ($g = 9.2 \text{ m/s}^2$). Determine the satellite's velocity and its period.
The earth's radius is $6.37 \times 10^6 \text{ m}$.



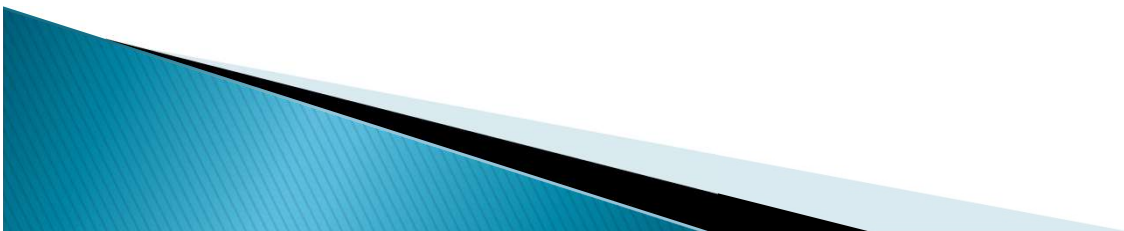
Problem

- ▶ A satellite is at an altitude $h = 200$ km above the earth ($g = 9.2 \text{ m/s}^2$). Determine the satellite's velocity, its period and angular velocity. The earth's radius is $6.37 \times 10^6 \text{ m}$.

Magnitudes only!

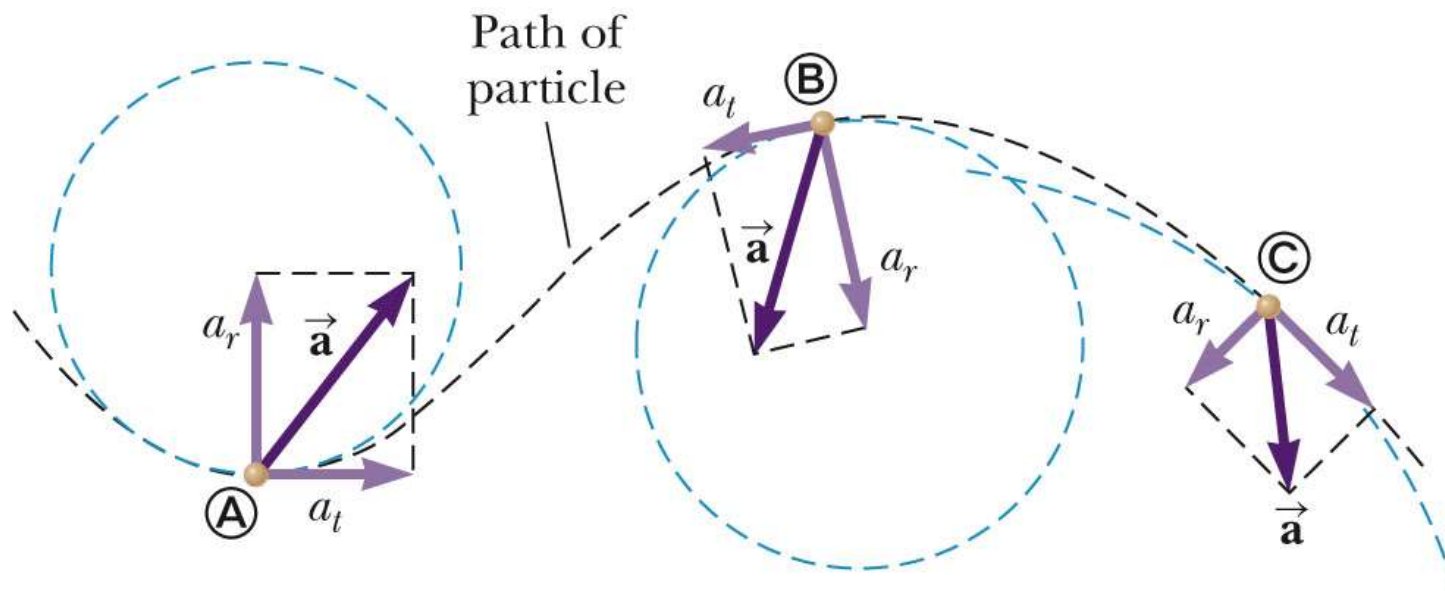
Certainly not geostationary!

How about in rad/hour?



Uniform Circular Motion

- ▶ The magnitude of the velocity could also be changing.
- ▶ In this case, there would be a *non-zero tangential acceleration*.
 - Note the changing acceleration vectors
- ▶ The motion would be under the influence of both tangential and centripetal accelerations.

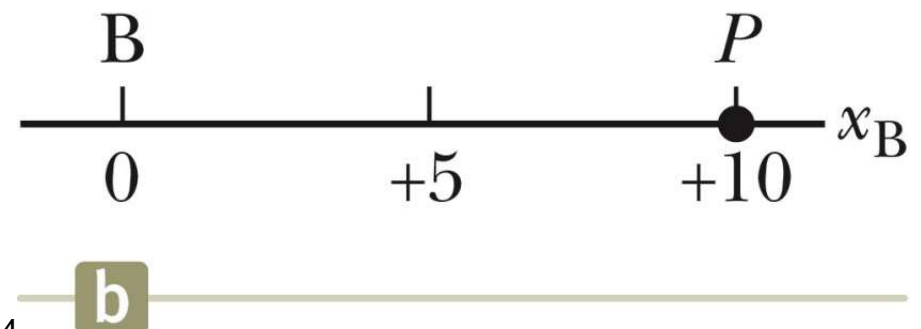
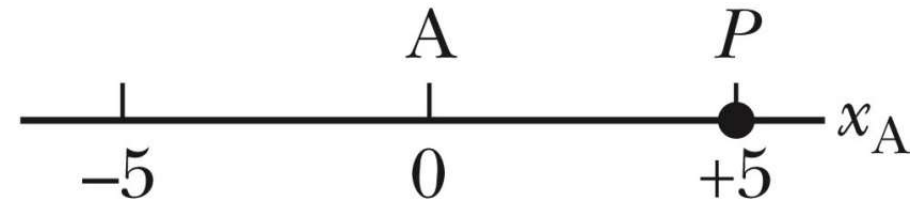
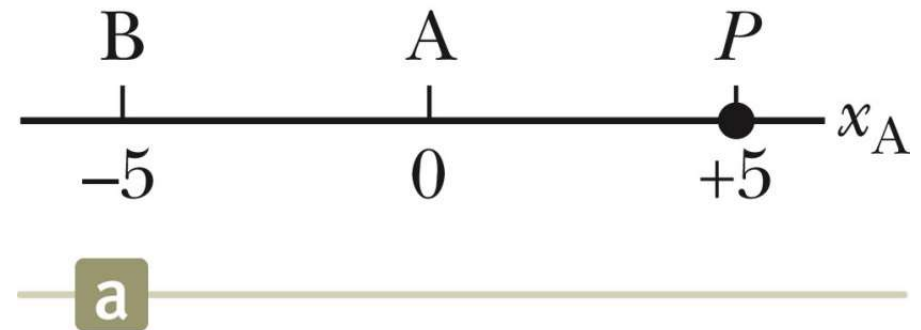


Relative Velocity

- ▶ Two **observers moving relative to each other generally** do not agree on the outcome of an experiment.
- ▶ However, the observations seen by each are related to one another.
- ▶ A frame of reference can be described by a Cartesian coordinate system for which an **observer is at rest with respect to the origin.**

Different Measurements example

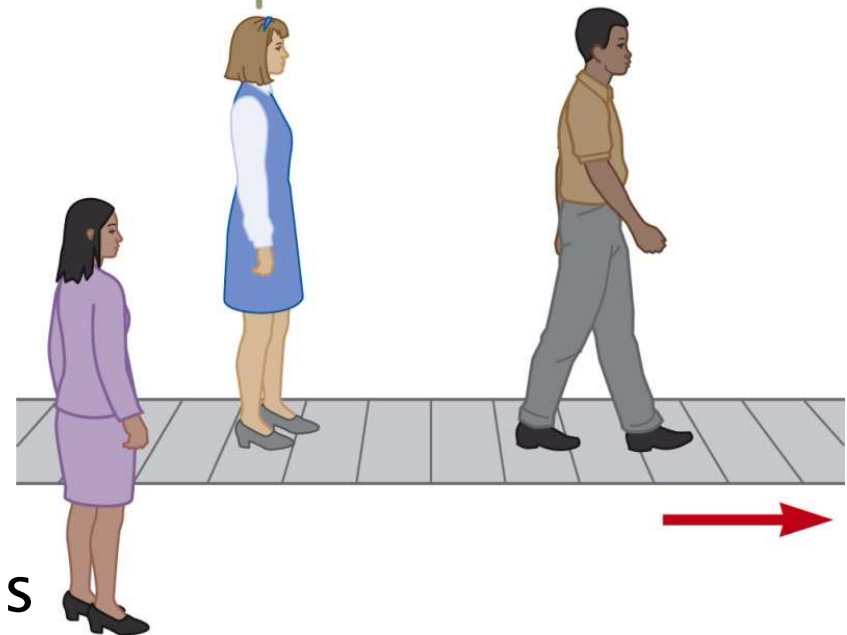
- ▶ Observer A measures point P at +5 m from the origin
- ▶ Observer B measures point P at +10 m from (B's) origin / cartesian frame of ref.
- ▶ The difference is due to the different frames of reference being used.
- ▶ What happens if B moves towards the origin of A?



Different Measurements example 2

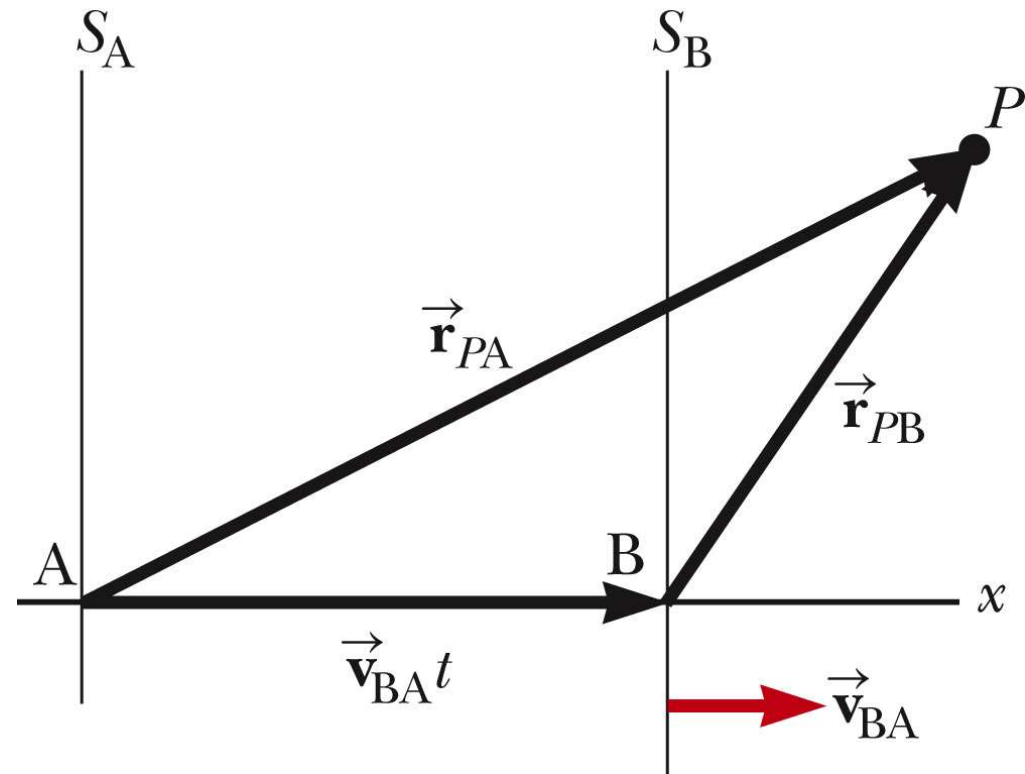
- ▶ The man is walking on the moving beltway.
- ▶ The woman on the beltway sees the man walking at his normal walking speed.
- ▶ The stationary woman sees the man walking at a much higher speed.
 - (A combination of the speed of the beltway and the walking).
- ▶ The difference is due to the relative velocities of their frames of reference.

The woman standing on the beltway sees the man moving with a slower speed than does the woman observing the man from the stationary floor.



Relative Velocity, generalized

- ▶ Suppose Reference frame S_A is stationary
- ▶ Reference frame S_B is moving to the right relative to S_A at $\vec{v}_{BA}$
 - This also means that S_A moves at $\vec{v}_{AB} = -\vec{v}_{BA}$ relative to S_B
- ▶ Define time $t = 0$ as that time when the origins coincide
 - ▶ At time t their positions are separated by $v_{BA}t$



$$\vec{r}_{PA} = \vec{r}_{PB} + \vec{v}_{BA}t$$

Notation

- ▶ The first subscript represents what is being observed.
- ▶ The second subscript represents who is doing the observing.
- ▶ Example $\vec{v}_{BA}$
 - The velocity of B (and attached to frame S_B) as measured by observer A

Relative Velocity, equations

- ▶ The positions as seen from the two reference frames are related through the velocity

$$\vec{r}_{PA} = \vec{r}_{PB} + \vec{v}_{BA}t$$

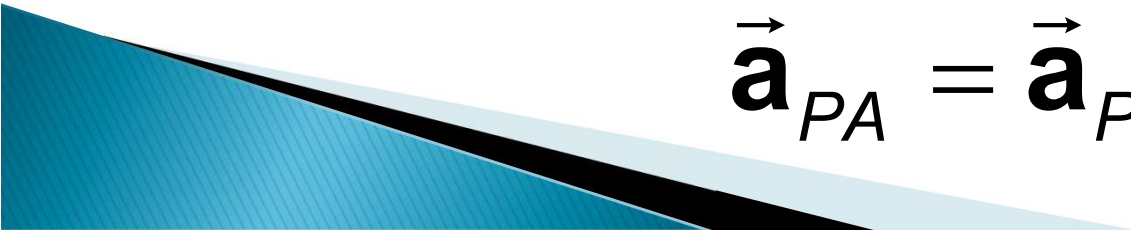
-

- ▶ The derivative of the position equation will give the velocity equation

$$\vec{u}_{PA} = \vec{u}_{PB} + \vec{v}_{BA}$$

- $\vec{u}_{PA}$ is the velocity of the particle P measured by observer A
- $\vec{u}_{PB}$ is the velocity of the particle P measured by observer B

- ▶ These are called the Galilean transformation equations.

$$\vec{a}_{PA} = \vec{a}_{PB}$$


Acceleration in Different Frames of Reference

- ▶ The derivative of the velocity equation will give the acceleration equation.
- ▶ The acceleration of the particle measured by an observer in one frame of reference **is the same as that measured by any other observer moving at a *constant velocity* relative to the first frame.**

Acceleration, cont.

- ▶ Calculating the acceleration gives

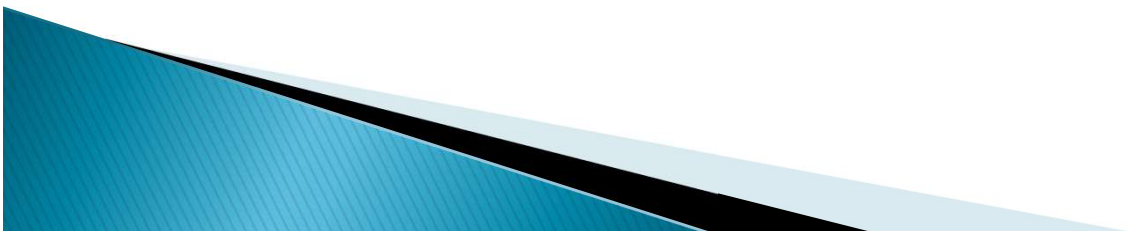
- ▶
$$\frac{d\vec{u}_{PA}}{dt} = \frac{d\vec{u}_{PB}}{dt} + \frac{d\vec{v}_{BA}}{dt}$$

- ▶ Since $\vec{v}_{BA}$ is constant, $d\vec{v}_{BA}/dt = 0$

- ▶ Therefore, $\vec{a}_{PA} = \vec{a}_{PB}$

Problem

- ▶ Suppose that Bob's velocity relative to Alex is a constant $v_{BA} = 52 \text{ km/h}$ and car P is moving in the negative direction of the x axis. (a) If Alex measures a constant $v_{PA} = -78 \text{ km/h}$ for car P , what velocity v_{PB} will Bob measure? (b) If car P brakes to a stop relative to Alex in time $t = 10 \text{ s}$ at constant acceleration, what is its acceleration a_{PA} relative to Alex? (c) What is the acceleration a_{PB} of car P relative to Bob during braking?



Problem

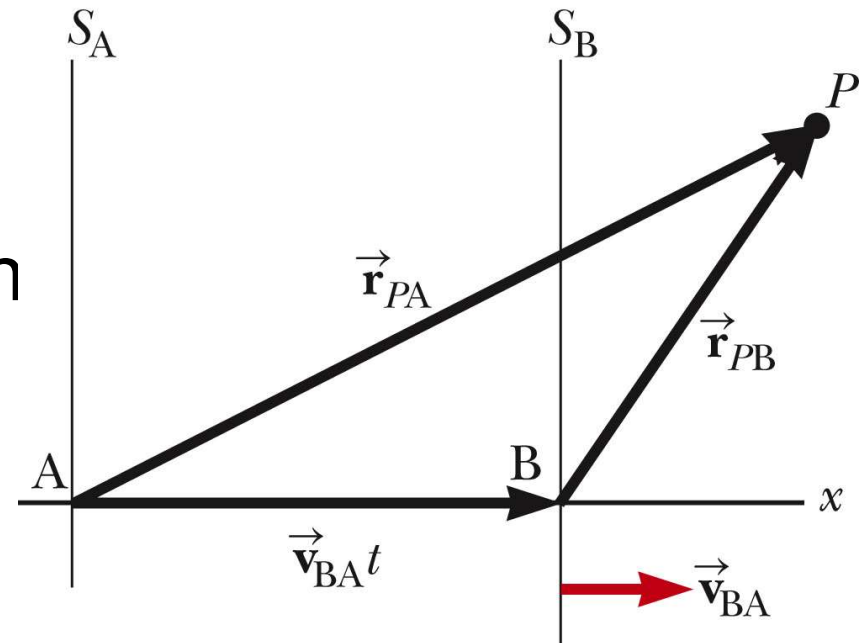
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Calculating Initial velocities

$$\begin{aligned}\text{(a)} \quad v_{PA} &= v_{PB} + v_{BA} \\ -78 \text{ km/h} &= v_{PB} + 52 \text{ km/h} \\ v_{PB} &= -78 \text{ km/h} - 52 \text{ km/h} \\ &= -130 \text{ km/h}\end{aligned}$$



Problem

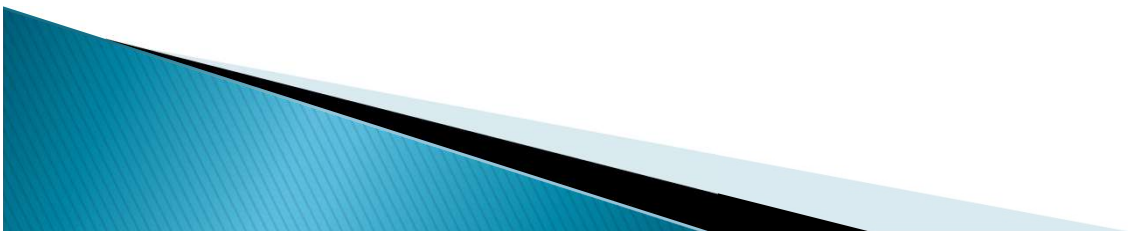
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(b) $v = v_0 + at$

$$a = \frac{v - v_0}{t} = \frac{[0 - (-78)] \frac{\text{km}}{\text{h}} \left(\frac{1000 \frac{\text{m}}{\text{km}}}{3600 \frac{\text{s}}{\text{h}}} \right)}{10 \text{ s}}$$
$$= 2.2 \text{ m/s}^2$$

Problem

- Suppose that Bob's velocity relative to Alex is a constant $v_{BA} = 52 \text{ km/h}$ and car P is moving in the negative direction of the x axis. (a) If Alex measures a constant $v_{PA} = -78 \text{ km/h}$ for car P , what velocity v_{PB} will Bob measure? (b) If car P brakes to stop relative to Alex in time $t = 10 \text{ s}$ at constant acceleration, what is its acceleration a_{PA} relative to Alex? (c) What is the acceleration a_{PB} of car P relative to Bob during braking?



Problem

A plane moves due E while the pilot points the plane somewhat S of E, toward a steady wind that blows to the NE. The plane has velocity $\vec{v}_{PW}$ relative to the wind, with an airspeed, relative to the wind of 215 km/h directed at angle θ S of E. The wind has a velocity $\vec{v}_{WG}$ relative to the ground with a speed 65 km/h, directed 20.0° E of N. What is the magnitude of the velocity $\vec{v}_{PG}$ of the plane relative to the ground, and what is θ .

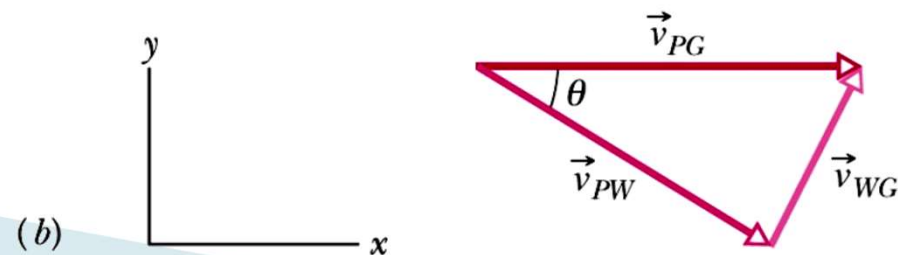
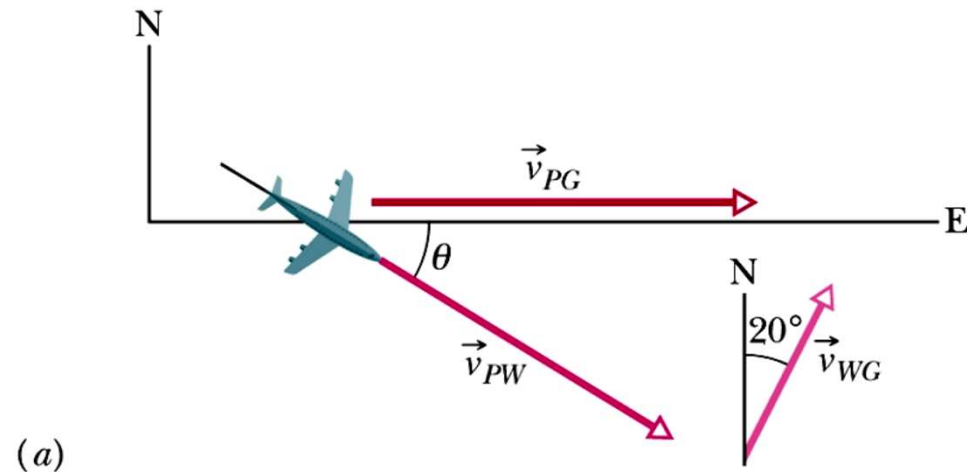
$$\vec{v}_{PW} = 215 \text{ km/h}$$

at some angle θ S of E

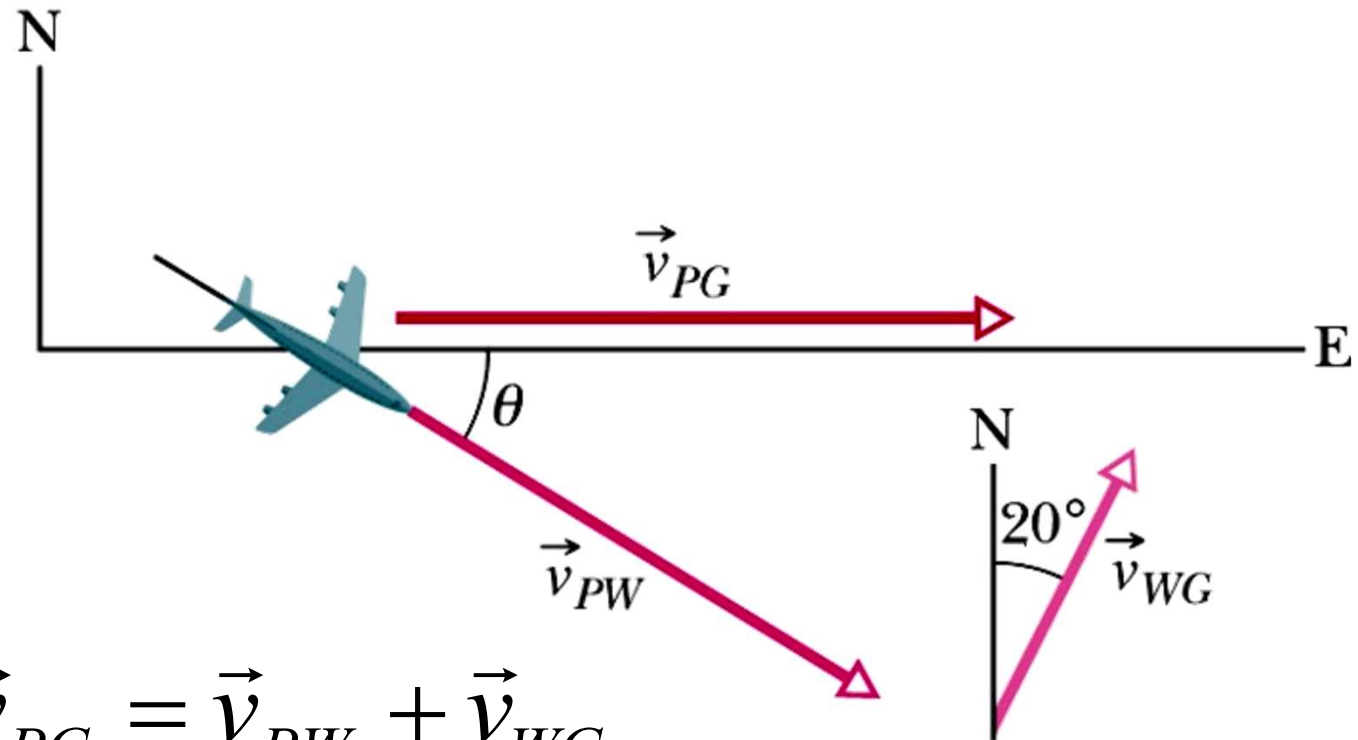
$$\vec{v}_{WG} = 65 \text{ km/h } 20.0^\circ \text{ E of N}$$

Determine vel of plane,

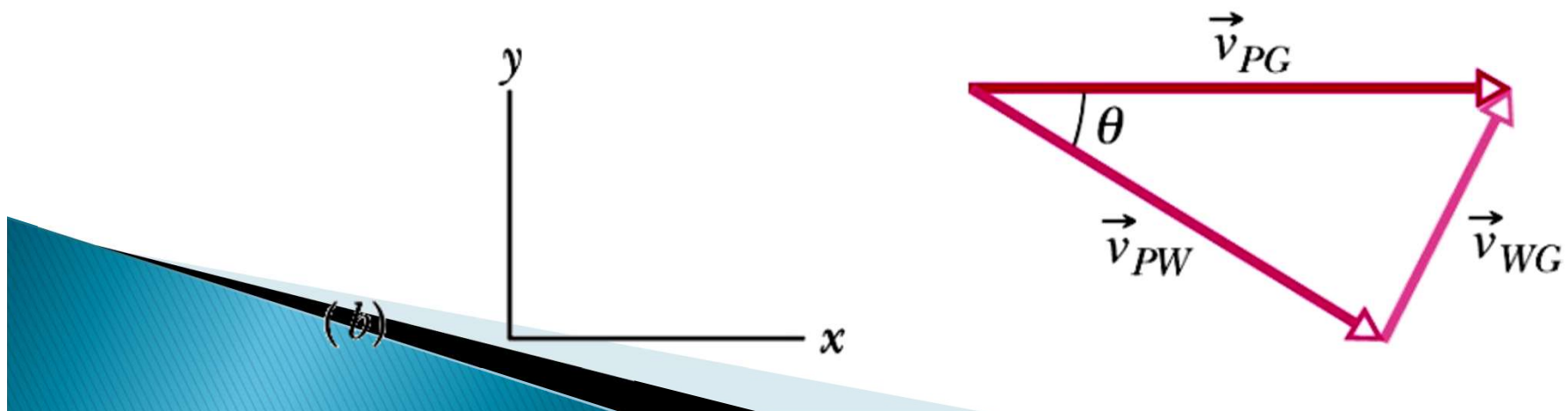
$\vec{v}_{PG}$, relative to the ground



Problem



$$(a) \quad \vec{v}_{PG} = \vec{v}_{PW} + \vec{v}_{WG}$$



Problem

A plane moves due E while the pilot points the plane somewhat S of E, toward a steady wind that blows to the NE. The plane has velocity $\vec{v}_{PW}$ relative to the wind, with an airspeed, relative to the wind of 215 km/h directed at angle θ S of E. The wind has a velocity $\vec{v}_{WG}$ relative to the ground with a speed 65 km/h, directed 20.0° E of N. What is the magnitude of the velocity $\vec{v}_{PG}$ of the plane relative to the ground, and what is θ .

STUDY Eg 4,8

